

Beyond the Signals

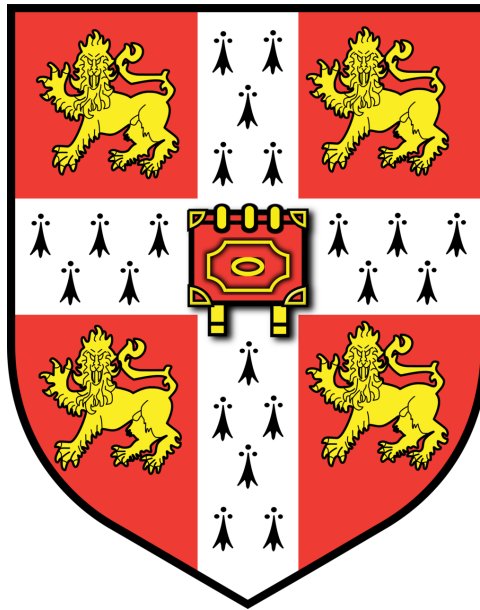
Evaluating the Signal Selection Test using simulated data

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Preface

This dissertation is the result of my own work and includes nothing which is the outcome of work done in collaboration, except where specifically indicated in the text. I do not analyse or generate any datasets, because this work proceeds within a theoretical and simulated approach. The dissertation does not exceed the word limit stipulated by the Degree Committee for Archaeology, Anthropology and Sociology.

Abstract

This study evaluates the reliability and scope of the Signal Selection Test (SST) for detecting selection in cultural time-series data under simulated archaeological conditions, focusing on its core statistic, the Frequency Increase Test (FIT). To emulate unbiased, content-biased, and conformist-biased transmission, I develop two simulation frameworks: a *snapshot* model that records every generation, and a *time averaging* model that aggregates generations into discrete time bins. By systematically varying population size, innovation rate, time series length, time averaging windows, and content and conformist biases, I aim to: (i) quantify the true-negative rate under neutrality, (ii) identify the conditions that maximise statistical power, and (iii) detect the conditions that minimise the proportion of non-analysable variants. Our results show that after appropriate calibration: (i) the SST correctly detects neutrality for more than 95% of the variants under both recovery conditions; (ii) statistical power only surpasses a sufficiently strong proportion under content bias when populations are sufficiently large and time is averaged, but remains consistently weak for conformist biased transmission; and (iii) time averaging improves SST's overall performance across all models except conformist transmission. We discuss implications for empirical studies of archaeological data, and offer practical recommendations for real-world data.

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Chapter 1. Introduction

Archaeologists have long sought to uncover the processes that shape cultural change over time, whether it's through artefact typologies and decoration, or through spatiotemporal reconstruction. To achieve this, many have adopted an evolutionary theoretical framework which envisions culture change as an interaction between functional adaptation and stylistic variation (Dunnell, 1978, 1980). However, discriminating between function and style is not a simple task, and it has required borrowing methods like neutrality tests directly from population genetics. Neutrality tests are based on the Wright-Fisher model of neutrality, where the variability of alleles is explained solely by stochasticity (Wright, 1931; Cavalli-Sforza and Feldman, 1981; Kimura, 1983; Boyd and Richerson, 1985). This cross-fertilisation yielded many insightful findings (Henrich, 2001; Neiman, 1995; Shennan and Wilkinson, 2001). However, these methods assume dynamics of idealised populations, which greatly differ from the fragmentary nature of the archaeological record (Madsen, 2012; Premo, 2014). For this reason, the uncritical application of neutrality tests risks generating ambiguous answers, especially when testing for selection events (Steele *et al.*, 2010; Crema, Kandler and Shennan, 2016; Crema, Bortolini and Lake, 2024).

Within this context, the Signal Selection Test (SST hereafter), developed by Feder, Kryazhimskiy and Plotkin (2014) for genetic time-series data, emerges as a promising technique. Unlike many neutrality tests SST is computationally efficient and explicitly designed to detect selection through the analysis of frequency changes through successive time points. Its potential for application to diachronic archaeological data seems clear, yet its suitability for the unique depositional realities of archaeological assemblages remains virtually untested.

This project confronts this critical methodological gap. It seeks to deconstruct the performance of the Signal Selection Test not on ambiguous empirical data, but within a controlled and simulated approach where the underlying processes are explicitly defined. By building this methodological approach we aim to systematically evaluate the test's performance and limitations. We are not only interested in questions such as “does it work?”, but how and when it works and what its failures reveal about the challenges of archaeological inference. Thus, we propose three core questions:

- How reliable is the SST under conditions of true neutrality, that is when all variants are replicated stochastically under realistic archaeological data structures? Before diving into the signals of selection, we must first determine the test's propensity for Type I error (false positives)
- How statistically powerful is the SST at detecting selection when it actually occurs? This involved assessing the probabilities that we incur in Type II error (false negatives), failing

to reject a selection event when it is present. Specifically, we model content-biased and conformist-biased transmission, both as reflections of selective advantages, one through quality and the other through quantity.

- How does the temporal structure for archaeological data affect the test's performance? An archaeological assemblage is an often complex system which entails multiple dynamics, including those involved in the process of deposition, but also in the nature of the artefacts itself. Through the long process of use, abandonment, re-use, deposition, and post-deposition we lose a lot of contextual information that lets us decode the cultural contents. By comparing the idealised *snapshot* scheme where every generation is recorded, against *time averaged* scenarios where time steps are aggregated into discrete bins—thus emulating archaeological conditions—we aim to isolate and critically evaluate the effects of assemblage accumulation on the output.

These questions are first framed in Chapter 2 within the broader context of the literature review, delving into the long-standing debates surrounding neutrality and selection tests in archaeology. We critically examine previous applications of neutrality tests, their caveats and recent alternatives, paving the way towards the methodological niche where this project is inserted. In Chapter 3 we review the cultural transmission models and time schemes to be simulated and describe experimental design. The agent-based simulations are divided into two experiments: Experiment 1, unbiased transmission, and Experiment 2 content and conformity biased transmission, each undergoing baseline and parameter sweeps and *snapshot* and *time averaging* recovery conditions. Subsequently, in Chapter 4 we outline the most insightful results from the experiments and discuss the undergoing theoretical processes that potentially explain them. Finally, after establishing a central thread between the results and the driving questions, we provide concluding remarks to seal this investigation, putting forward potential pathways to explore in future research.

Chapter 2. Literature review

Archaeology has long been studied under the evolutionary paradigm. Multiple attempts have sought to apply the Darwinian principles to explain how culture has changed throughout its long history. As thorough as many were, no such attempts came close to the types of predictions that evolutionary models proposed during the so-called "Modern Synthesis". Geneticists found a way to broadly describe how patterns observed in populations are shaped by mechanisms observed at the individual level (Dobzhansky, 1937). These patterns could thus be dichotomised into neutral and selective: the former yields no correlation with the environment, relying solely on stochasticity, and the latter describes the either positive or negative effect of a trait in association to its environment.

Many archaeologists came close to this dichotomy, but it was the seminal paper published in 1978 by Robert C. Dunnell that first posed the theoretical ground for evolutionary archaeology to bloom. In the cited article, Dunnell proposes an evolutionary approach which, analogously to genetic population models, entails two types of elements: *style*, the type of variation that has no selective value, and *function*, as the form of variation that alters the fitness of those populations in which occur (1978, p. 199). Both types yield multiple outcomes observable in the archaeological record, outcomes which in many occasions are not easily distinguishable from each other.

Since the establishment of this dichotomy, evolutionary archaeologists have been interested in precisely distinguishing cultural patterns that arise from random processes from those driven by selective forces (O'Brien and Holland, 1990; Neiman, 1995; O'Brien and Lyman, 2000; Lipo, 2001; Shennan and Wilkinson, 2001). To achieve this, researchers have borrowed a powerful suite of tools from population genetics known as neutrality tests. A neutrality test fundamentally assesses if the observed frequency and diversity of cultural variants deviates from expectations under a purely unbiased transmission model (Madsen, 2012). Examples include the prevalence of certain pottery motifs and handaxe morphology (Neiman, 1995; Kempe, Lycett and Mesoudi, 2012). This null model assumes change arises only through blind copying errors (drift) and occasional novel inventions (mutation) (Eerkens and Lipo, 2005).

Evaluating the performance of these tests is critically important because the archaeological record is not an ideal sample. It is often fragmented, representing only a fraction of the past material culture; and most critically, it is time-averaged, meaning it aggregates materials deposited over many generations (Premo, 2014). These methodological issues raise significant questions about the reliability of the inferences provided by neutrality tests, creating a need for methods tailored to the realities of archaeological data (Kandler and Crema, 2019).

2.1. Foundational Studies: Applying neutrality theory to culture

The initial wave of research was motivated by a foundational question: Can we explain the patterns of culture change seen in the archaeological record without invoking complex selective forces? The first years of theoretical development were characterised by the publications by Robert C. Dunnell, who equated the archaeological concept of “style” with selectively neutral variation, building a powerful bridge between culture history and Darwinian evolution (1978, 1980). In the following years two seminal works put forward the formal models of cultural transmission where unbiased transmission acted as the null, with selection/biased models pivoting around (Cavalli-Sforza and Feldman, 1981; Boyd and Richerson, 1985). Although these were not explicitly tailored to the archaeological reality, many ensuing studies would benefit from borrowing these models.

An early application by Fraser Neiman (1995) demonstrated through simulations and empirical data that the classic “battleship curves” assumed in seriation models could be generated by a neutral process of random copying and innovation, often called “cultural drift”. This work established a solid testable baseline for explaining cultural change, to the extent that many researchers directly applied Neiman’s approach to empirical data after him. For instance, Shennan and Wilkinson (2001) analysed the LBK Merzbach pottery dataset using theoretical vs empirical diversity estimates, following Ewen’s sampling distribution (Ewens, 1972). They concluded that variation within the assemblage studied could not be defined dichotomously, but rather within a broader spectrum of possibilities between drift and selection (p. 592). In a contrasting landmark study, Kohler et al. (2004) found that changes in ceramic diversity found on the Pajarito Plateau contrast significantly with expectations given by Neiman’s neutral model (p. 115). In fact, the low levels of diversity were interpreted as strong evidence for conformist transmission—a form of social learning bias where individuals preferentially copy the most common variants (Boyd and Richerson, 1985).

Another relevant empirical contribution is that done by Steele, Glatz and Kandler (2010), who studied Hittite bowls to perform quantitative hypothesis testing against the neutral model. The authors concluded that although frequency distribution of rim sherds aligns with neutral expectations, a deeper analysis reveals that functional characteristics—like vessel diameter—correlate with abundance, perhaps indicating latent selective pressures (p. 1357). Therefore, neutrality tests are only as good as the traits being analysed, and functional equivalence cannot be assumed (Steele, Glatz and Kandler, 2010).

These studies primarily relied on statistical tools like the Ewens-Watterson test using the sample homozygosity and Slatkin’s “exact” test (Ewens, 1972, 2004; Slatkin, 1994, 1996). These tests evaluate the shape of frequency distributions against the mathematical expectations of neutrality—a population at mutation-drift equilibrium (Ewens Sampling Distribution). The main difference between tests is that the latter conditions on both sample size (n) and observed allele count (k) to assess the full frequency spectrum of the population; whereas the former conditions only on sample size (n) and tests neutrality via homozygosity (see Madsen (2012) for a more detailed comparison, and Kandler and Crema, (2019) for a review of theory and application of both tests). These techniques have become standard methods for comparing diversity estimates, hypothesis testing against neutrality, and population dynamics (Neiman, 1995; Lipo *et al.*, 1997; Rorabaugh, 2014).

Even historical cultural datasets have been studied through neutral hypothesis testing, such as changing popularity of baby names, purebred dogs and US citation patents (Bentley, Hahn and Shennan, 2004; Acerbi and Bentley, 2014). The apparent positive results that these studies yielded can translate into a good permeability of neutrality tests to other disciplines—a vigorous property of

Darwinian theory.

2.2. Methodological Struggles: The reality of archaeological data

After this stage of cross-fertilisation between genetic and culture theory, archaeologists were driven by methodological questions regarding the nature of the archaeological record. Neutrality tests had showed insightful results for archaeological data, but many issues such as spatial and time complexity, and social learning errors remained unexplained (Bentley, Hahn and Shennan, 2004; Eerkens and Lipo, 2005; Shennan, 2011; Madsen, 2012; Premo, 2014; Kandler and Shennan, 2015).

The work of Madsen (2012, 2020), Premo (2014) and Porčić (2015) provided a stark answer by directly confronting the time averaging problem. Through a simulated approach they demonstrated a critical flaw: time averaging systematically inflates richness and flattens evenness distributions within assemblage, causing an increase in Type I errors. In other words, the very formative process of the archaeological record can make neutral variants appear non-neutral. Porčić (2015) provided evidence that assemblage accumulation yields significant noise in diversity estimations in conformist and anti-conformist models (p. 1089)—similarly to neutral models as shown by Madsen (2012) and Premo (2014).

Furthermore, some authors tackled the issue of spatial complexity, demonstrating that spatial scale can have a great impact on diversity measures, and in extension forms of cultural transmission (Lipo *et al.*, 1997; Lipo, 2001; Premo and Scholnick, 2011). For instance, Premo and Scholnick (2011) showed through agent-based simulations how highly localised populations following an unbiased learning rule could increase the number of rare variants (“singletons”), resulting in statistical signals that could be potentially confused with biased transmission (p. 175). This added another layer of ambiguity to the interpretation of tests results from potentially structured populations. Perhaps this is not that much relevant to this study

This is often mentioned in the literature as “the principle of equifinality”: different social and non-social learning processes lead to coinciding outcomes (Premo, 2010; Rorabaugh, 2014; Barrett, 2019). For instance, a pattern of low artefact diversity could result from a behavioural process like conformist transmission, but it could also stem from a demographic bottleneck-like event or functional constraints. Conversely, high diversity may result from anti-conformist behaviour, or simply a high rate of innovation or individual experimentation in an unbiased context (Rorabaugh, 2014). Because such dynamics blur the signals that distinguish one process from another, outcomes from neutrality tests become uncertain even when results are clear (Madsen, 2020). And to add another layer of complexity, the long array of possible alternative hypotheses to neutrality (e.g. social learning biases) reduces the information yielded by null-rejection (Shennan and Wilkinson, 2015).

2001).

While equifinality complicates the interpretation of observed patterns, measuring artefact diversity rests on a critical assumption about what constitutes a cultural unit. This assumption is far from secure, leading to a persistent challenge in cultural evolution: defining the units of transmission (Eerkens and Lipo, 2005; O'Brien and Shennan, 2010). Cultural information lacks the discrete and observable units found in genetics. Furthermore, archaeologists study the material *products* of transmission (empirical units), which are often far from the mental *recipes* (ideational units). Crema, Bortolini and Lake, (2024) have put forward the argument that there is often a mismatch between the frequency distribution of mental templates and objects, which potentially leads to inferential biases.

2.3. The Path Forward: Developing more robust methods

In response to this methodological impasse, the most recent research has shifted toward developing more robust and context-aware inferential frameworks. These move beyond simple null-hypothesis testing and are better suited to the complexities of archaeological data. In fact, they seek to build more realistic models of culture change, instead of simply asking whether selection occurred or not.

One of the most promising approaches is the “generative inference approach”, put forward by Kandler and Powell (2015, 2018). This framework uses simulation and Approximate Bayesian Computation (ABC) to generate thousands of datasets under multiple competing transmission schemes. Through a quantitative comparison of simulated outcomes and the observed archaeological patterns, researchers can determine which underlying model provides the most likely explanation, thus confronting equifinality (Kandler and Crema, 2019). The ABC was explored by Crema, Kandler and Shennan (2016), who compared the LBK Merzbach pottery distribution with a simulated model of cultural transmission comprised of one parameter that describes a range from conformist to anti-conformist, paired with unbiased transmission. Although failing to find matches between empirical and simulated data, the authors hinted at a more complex non-equilibrium model to find the best-fit (pp. 8-9).

In a complementary vein, other methods aim to disentangle the forces of evolution more directly. The Price Equation (Price, 1970), for instance, partitions cultural change into two key components: changes correlated with payoffs or fitness (covariance term) and changes uncorrelated with payoffs (expectation term)—thus adjusting to the dichotomy of functional/selective and stylistic/neutral variation (Henrich and Boyd, 2002; Brantingham, 2007). The insightful feature of this model is that it allows to quantify the relative strength of both selective and stochastic forces operating simultaneously. The application of this method by Brantingham and Perreault (2010) to

Hittite bowls provided evidence of selection forces acting predominantly in favour of certain rim types, coinciding with Steele, Glatz and Kandler's conclusion (2010).

2.4. Selection Tests: The Signal Selection Test

As much as these techniques have helped archaeologists infer potential patterns of selection, they suffer from two key limitations. First, some like the ABC rely on complex simulations and require large datasets that make them computationally intensive and potentially difficult to apply to the archaeological record. Secondly, they are indirect, meaning they don't detect selection itself but only a statistical departure from neutrality, nor give an estimated measure of selection for specific variants (Kandler and Shennan, 2013). This highlights the need for robust, accessible and statistically powerful tools that operate effectively with time-series data. One such method is the Signal Selection Test (SST) and its main statistical resource, the Frequency Increase Test (FIT).

Developed by Feder, Kryazhimskiy and Plotkin (2014), it is a statistical test designed to identify signals of selection in genetic time series data, by analysing allele changes between time steps. One of the main advantages which posits this tool forward is that it is computationally more efficient than ABC approaches, avoiding the need for intensive simulations to establish significance (for a more detailed explanation see Chapter 3).

FIT has seen little to no application in cultural evolution, especially in archaeology. It was notably adapted for culturomics data by Newberry et al. (2017) yielding insightful results on language evolution. Moreover, it was briefly benchmarked against other software tools by Vlachos et al. (2019). And the only application to archaeological data is available at a public repository (Marwick, Bacovcin and Kryazhimskiy, 2019), consisting of an analysis of the LBK Merzbach Valley pottery decorative motifs (Fig. 1). As useful as it is as a coding resource, it does not explore the test's scope and limitations in great depth. Thus, its suitability and performance for archaeological conditions remain unevaluated. It is to be considered if the test detects both neutrality and selection when confounding processes such as time averaging and equifinality are at play.

Precisely, this project is situated at this critical intersection of methodological need and opportunity. It seeks to (i) systematically evaluate the performance of the SST as a tool for detecting selection, (ii) identify its limitations, and (iii) establish its viability as a practical and accessible tool for archaeological datasets. In the next chapter we describe its core statistical properties and the methodological approach, which we advance that consists in a series of controlled experiments using simulations of cultural transmission regimes, combined with multiple time schemes and parameter combinations.

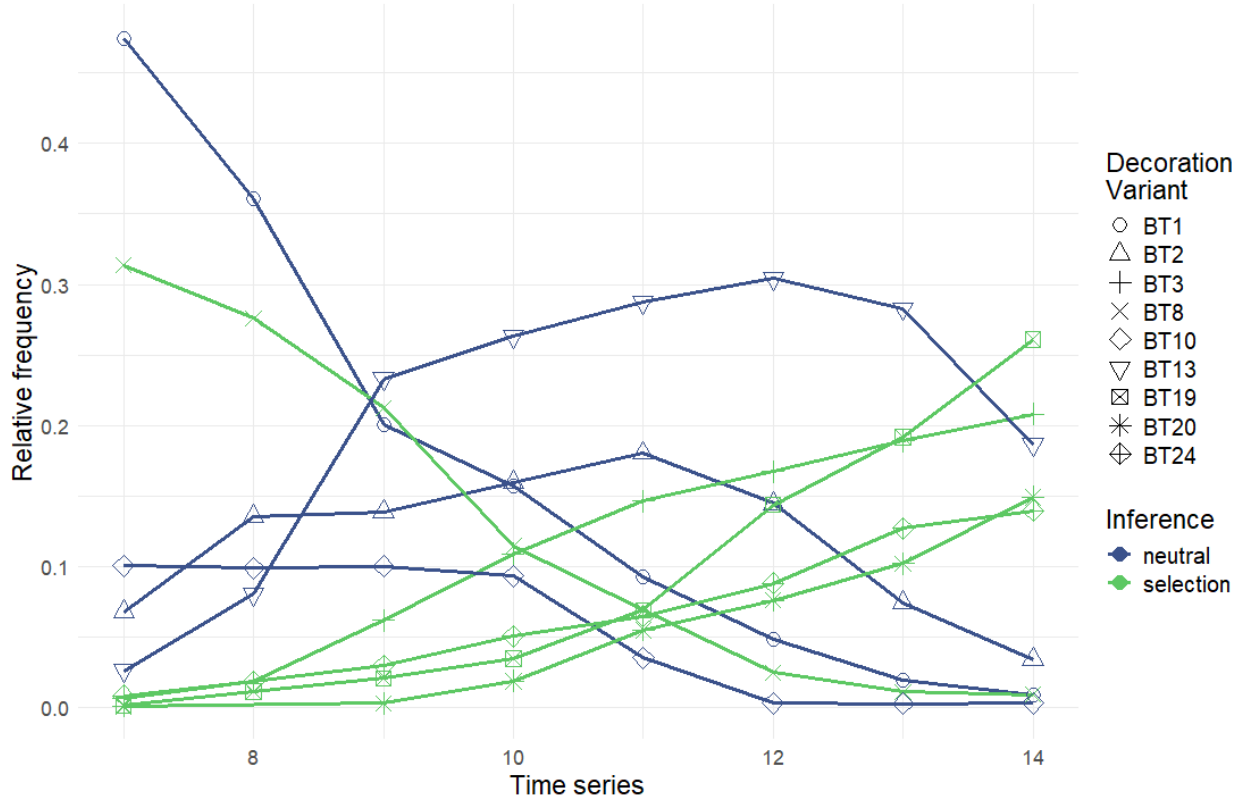


Figure 1: Application of the FIT to the LBK Merzbach Valley ceramic assemblage. Each point shape represents the frequency of a decorative motif variant at a single time step (see Boelicke et al., 1994 and Shennan & Wilkinson, 2001 for the complete reference of the variants). Variants detected as selected ($p\text{-value} \leq 0.05$) are shown in green, and variants detected as neutral ($p\text{-value} > 0.05$) are shown in purple. Slightly modified from <https://github.com/benmarwick/signatselect>, who conduct a FIT analysis.

Chapter 3. Methods

3.1. Signal Selection Test

The technique under evaluation is what Marwick, Bacovcin and Kryazhimsky (2019) called the Signal Selection Test (SST, or *signatselect* as the name of the R package), whose statistical machinery is principally comprised by the Frequency Increase Test (FIT). FIT is a statistical tool which aims to detect signals of selection in time series frequency data. It was originally designed to infer selection events in genetic time series data. Essentially, it employs a one-sample t -test to compare the rescaled allele frequency differences between two successive intervals, with theoretical expectations under a neutral Wright-Fisher evolutionary model (Feder, Kryazhimskiy and Plotkin, 2014; Vlachos *et al.*, 2019).

Significant deviation of observed mean frequency increases from zero results in rejection of

the null hypothesis ($\alpha = 0.05$), which posits neutral evolutionary dynamics driven by stochasticity alone. Consequently, the interpretation given rejection is that selection forces are at play at certain time steps (Feder, Kryazhimskiy and Plotkin, 2014) – hence the test’s designation. In other words, if the observed allele distribution cannot be adequately described by neutrality alone, the test flags statistically significant evidence of selection.

Additionally, the SST framework provides a set of outcomes under both neutral and selection hypotheses, which are crucial for characterising allele-frequency dynamics. The ones that have deserved more attention in the literature are the estimated population size (N) and coefficient of selection (s) (Feder, Kryazhimskiy and Plotkin, 2014; Newberry *et al.*, 2017). Their importance resides in the fact that they both specify the likelihood function $L(\text{Data}; N, s)$ of allele frequencies (2014, p. 511), in conjunction with sampling noise and other nuisance parameters (Ewens, 2004). Essentially, both parameters describe the probability of observing allele frequencies transitioning from an initial state v_0 at time t_0 to a subsequent state v_{i+1} at time t_{i+1} , formally represented as

$$Ps(v_i, t_i | v_{i+1}, t_{i+1})$$

where $s = 0$ under the null and $s > 0$ under the selective advantage hypothesis (Feder, Kryazhimskiy and Plotkin, 2014, eqs. 9–10).

Crucially, the one-sample t -test compares two successive frequency differences, which requires at least three temporally ordered observations—an initial, an intermediate, and a final time point—to generate the paired frequency increments. This is what we refer to as the “three-time-point rule”, whose implications we discuss in chapter .

The core analytical step of the test involves computing a log-likelihood ratio (LLR) which compares the maximised likelihood of the data under the selection model, to that under neutrality. This ratio is then log-transformed to obtain a test statistic that quantifies the strength of evidence for selection. The LLR is then converted into a t -statistic from which a two-tailed p -value is derived (Feder, Kryazhimskiy and Plotkin, 2014, eq. 11), with a large ratio indicating that the data are better explained by the selection model. In other words, we obtain a statistic that directly measures how much better the selection model explains the data. In Table 1, the column labelled “LL” reports the log-likelihood ratio, highlighting, for example, that BT20’s trajectory is most likely explained by a positive-selection model.

Although FIT also computes an estimated initial frequency of the allele (f_0) in the sampled time series (Marwick, Bacovcin and Kryazhimsky, 2019) – which has not been acknowledged in the literature –, our analysis here focuses on evaluating the first outcome of the test: p -values

and statistical significance. Specifically, its capacity to detect selection in cultural evolutionary time series affected by archaeological conditions. We do not explore s estimations because our model of selection is built differently from Feder, Kryazhimskiy and Plotkin (2014) to suit cultural evolution conditions (although borrowed from genetic evolutionary models (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981)). Moreover, the referenced researchers already performed a power analysis on s estimations and compared results with genetic data.

Given that FIT assumes Wright-Fisher-like dynamics and independent frequency increments (Feder, Kryazhimskiy and Plotkin, 2014), we critically assess whether the outcomes predicted from these models are convergent when artefacts are deposited and accumulated irregularly across depositional strata (Premo, 2014; Crema, Kandler and Shennan, 2016). By simulating multiple scenarios of cultural transmission and assemblage properties, we aim to assess quantitative measurements of the performance of the FIT applied to archaeological data. The test implementation and statistical analyses were conducted using a custom R script (version 4.5.0; R Core Team, 2025) based on the scheme of publicly available repos published by some of the authors referenced above (see “Appendix” below for access to code and scripts).

3.2. Models of cultural transmission

To gauge FIT’s statistical power within a simulation framework we introduce three individual-based models of horizontal cultural transmission in which social learning rules differ (Cavalli-Sforza and Feldman, 1981; Boyd and Richerson, 1985; Tomasello, Kruger and Ratner, 1993). Each model aims to emulate how cultural information spreads within a single population, thus shaping variant frequencies. We range from imitation by naïve individuals regardless of variants’ properties (neutral/unbiased transmission), to biased transmission of variants based on their qualitative (content/direct) and/or quantitative (conformist/anti-conformist) features (Boyd and Richerson, 1985; Henrich and Boyd, 1998; Acerbi and Bentley, 2014).

We describe a set of parameters and assumptions which are shared across models, aimed at describing the core structure of the simulated populations. Each one has a population size N , the number of discrete cultural artefacts at each time step, with an ID that represents the variant carried; innovation rate μ (analogous to mutation rate), the probability at a single time step that an individual abandons a socially learned variant in favour of a novel one¹, so that if $\mu = 0.1$ there is a 10 % chance that this event takes place; an unobserved burn-in period B , during which only random copying, drift, and random innovation operate. The burn-in stage minimises dependence

¹Here we assume an infinite allele model, where there is no chance of convergent innovation or mutations yielding to an existing or ever existed variant. All variants that are introduced are novel, and when they disappear they never reappear again.

on the arbitrary initial conditions by allowing the population to reach its stationary (equilibrium) distribution (Crema, Kandler and Shennan, 2016). It also mirrors the reality that archaeologists usually encounter assemblages at a point where formative dynamics have stabilised. Finally, the number of recorded transmission events t , or time steps that comprise the whole series, all of equal length.

It is worth noting that no analysis presented here fully accounts for equifinality, i.e. different conditions arriving at coinciding final states (Barrett, 2019) – although the matter is briefly discussed in Chapters 4 and 5. In this sense, we acknowledge that real-life data may not respond to constant conditions of cultural transmission, nor to non-overlapping social learning biases (Madsen, 2012). Nonetheless, this study aims not to cover the totality of cultural variation in the archaeological record. Instead, it introduces new insights and a recent statistical tool to explain some of this variation under idealised archaeological settings.

3.2.1. Neutral/Unbiased transmission

In the unbiased infinite-variant transmission model, naïve individuals randomly copy a variant from any member of the population with a probability $1 - \mu$, choosing a variant i in proportion to its current frequency:

$$f_i = \frac{n_i}{\sum_j N_j} = \frac{n_i}{N} \quad (1)$$

and it innovates with a probability μ , i.e. a novel variant introduced in the population—assuming the variant has never been present. Consequently, the probability that a learner adopts an existing variant i is:

$$P(i) = (1 - \mu)f_i \quad (2)$$

This model is borrowed from a haploid Wright-Fisher neutral model of evolution in which the frequency of a variant is a function of population size (N) and mutation rate (μ), here innovation rate (Kimura and Crow, 1964; Ewens, 2004) (Fig. 2).

In principle, N is the most important parameter in this model because it specifies the probabilities of transmission in two ways. When the total population is small, this proportionality amplifies an early advantage: a variant seeded at a high starting frequency f_0 , immediately captures $f_0 \times N$ learners, making it far more likely to persist and spread than a rarer variant (Boyd and Richerson, 1985). But what is crucial to our models is that population size determines the moment of

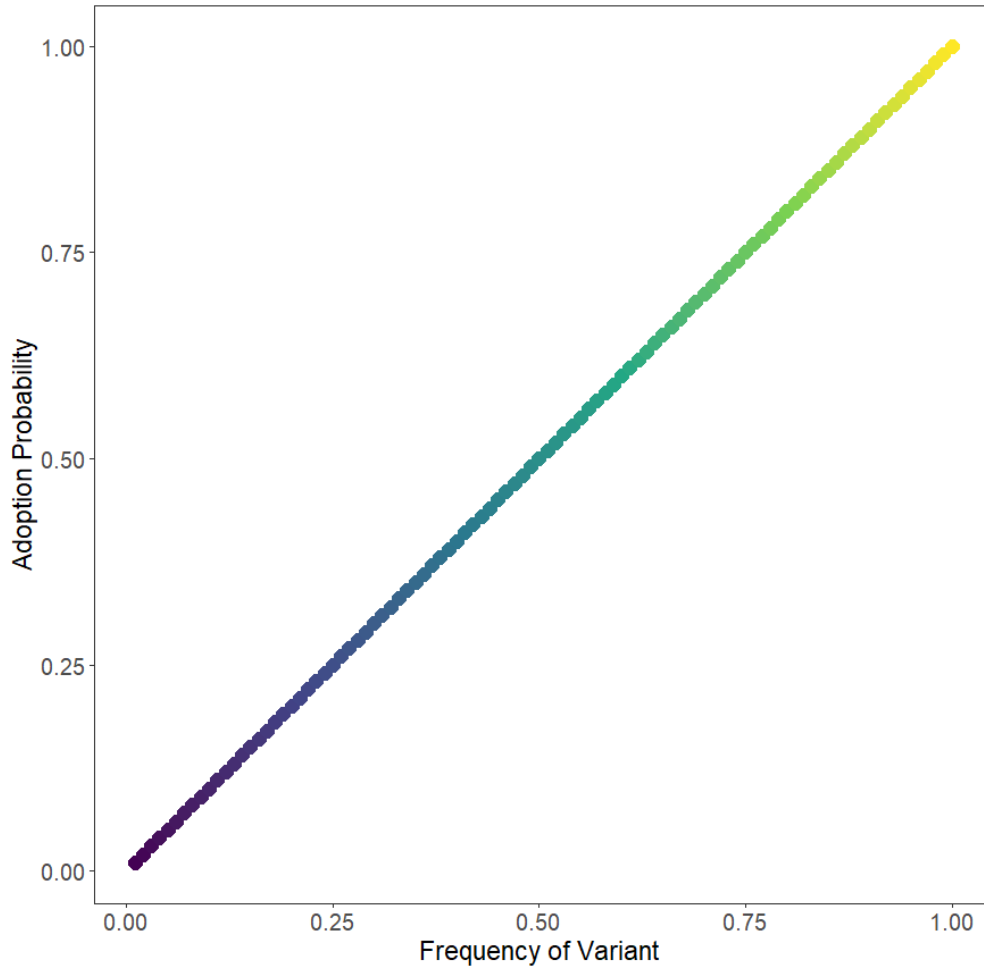


Figure 2: Adoption probability of a variant in an unbiased transmission scenario. The probability that an individual adopts an existing variant in their population is dependent on the frequency of that variant. This is the main premise of the haploid Wright–Fisher model of neutral evolution applied to cultural transmission theory: if we draw two learners at random, the probability they choose the same variant is $1/N$, and $1 - 1/N$ from a different model (Neiman, 1995).

fixation/extinction (Acerbi, Mesoudi and Smolla, 2022). Larger N will extend the fixation/extinction event, whereas smaller N will shorten this time to even a few generations (Fig. 3).

μ directly mirrors the mutation role in genetic models (Shennan & Wilkinson, 2001). Under the infinite-variant assumption, the expected number of coexisting variants at equilibrium is $\theta = 4N\mu$. Thus, when μ is low few new variants appear on average each generation, so the population has very low standing diversity. This accrues the effect of drift, as variants wander rapidly to fixation or extinction, especially in low N . Conversely, when μ is high many new variants are continually entering the population, maintaining high diversity even under small N (for further discussions on drift and interactions between N and μ in genetic models see Kimura, 1983, and archaeological

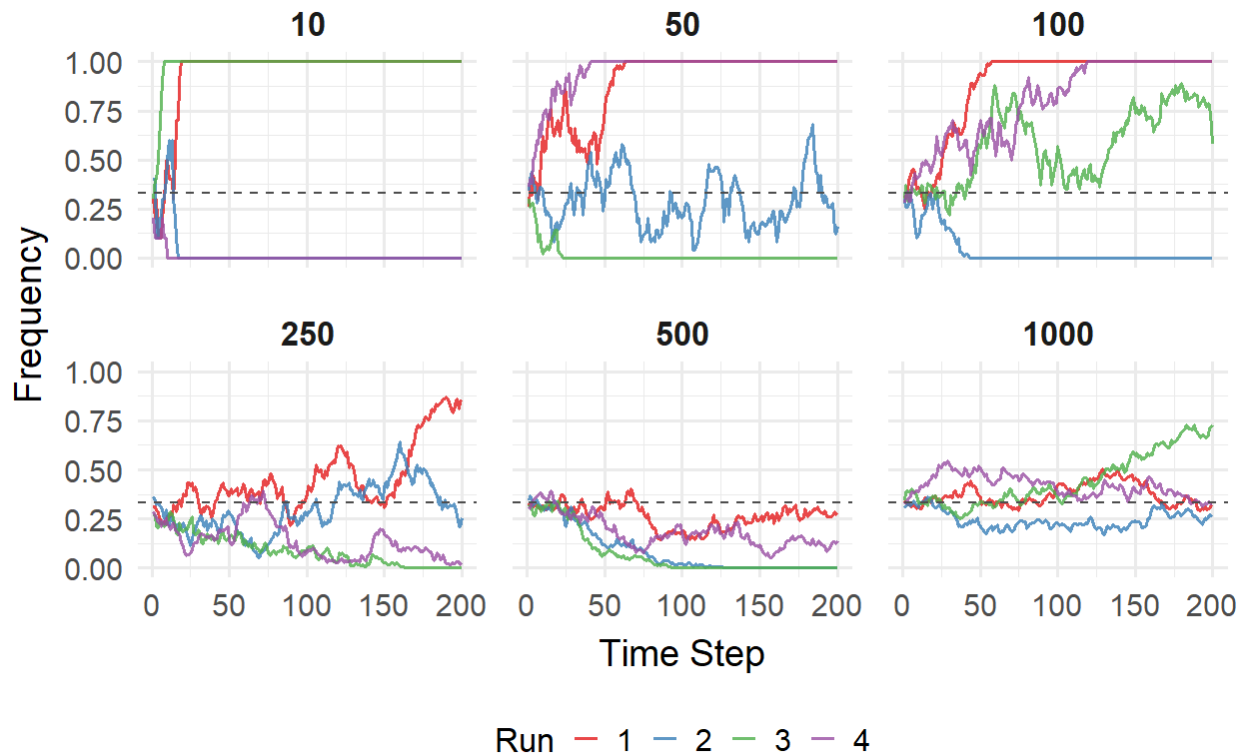


Figure 3: Simulated time series of a cultural variant under neutral (unbiased) transmission with equal initial frequencies at 0.33, and varying population sizes and $\mu = 0$.

applications Neiman, 1995).

Due to the stochastic nature of this model many authors have used it as the null hypothesis (Neiman, 1995; Shennan and Wilkinson, 2001; Steele, Glatz and Kandler, 2010), and so does the FIT (Feder, Kryazhimskiy and Plotkin, 2014; Newberry *et al.*, 2017). Unbiased transmission reflects a process of social learning that does not consider cost differences between variants – supposing individuals have access to this information. Thus, it has been suggested that there is no empirical evidence to support a homogeneous and purely unbiased copying strategy, meaning that all transmission is somewhat biased (Eerkens and Lipo, 2005; Madsen, 2012; Porčić *et al.*, 2024). This raises new questions regarding the applicability and sensibility of neutral models when archaeological data is under analysis.

3.2.2. Content biased transmission

In the content-biased model, each variant carries a “quality” score b , and learners adopt variants with probabilities proportional to their variance and the magnitude of b , introducing

directional selection into the copy/imitation process (Boyd and Richerson, 1985). Learners copy a demonstrator with a better chance if their variant is more “attractive”, measured by a score of $1 + b$ (Henrich and Boyd, 2002; Claidière and Sperber, 2007; Acerbi and Bentley, 2014). Adoption probability is described by equation 1, but this time we add the bias score similarly to how Boyd & Richerson expressed it (1985, eq. 5.4):

$$P(i) = (1 - \mu) \frac{f_i a_i}{\sum_{j=1}^k f_j a_j} \quad (3)$$

where a_i is the attractiveness score that varies the probability of adoption of the i th variant:

$$a_i = 1 + b_i, \quad \text{with } b_i \in [-1, \infty) \quad (4)$$

And the full extension of the model, where the weights are normalised by the denominator, so that the weight of the i th variant is calculated relative to the other models encountered by the individual (Boyd & Richerson, 1985, p. 140):

$$P(i) = (1 - \mu) \frac{f_i (1 + b_i)}{\sum_{j=1}^k f_j (1 + b_j)} \quad (5)$$

so that when $b = 0$ adoption probability is fully dependent on N and μ , thus being equivalent to equation 1. But when $b < 0$ or $b > 0$ transmission is biased, which translates into negative and positive selection towards differing-quality variants.

Other authors have modelled content bias by allowing individuals to adopt a variant when is more attractive than the one they carry—i.e. $a_j > a_i$ (Acerbi and Bentley, 2014). However, in our model we assume that individuals do not have all the qualitative information at their disposal, so they virtually cannot evaluate whether their variant is “better” than others to which they have access. Instead, we assume that they

An important feature of this model is that most variants are assumed to have $b = 0$, meaning they are transmitted neutrally. Only a small proportion of the population is affected by the content bias ($b > 0$). We reason that even if neutral variants experience a change in their probability, realistically there is only a small fraction that is under selected. Moreover, theoretically b is unbounded (4), although we intuitively argue that higher b values are more rarely seen.

The additional probability given by the bias strength can lead to significant increments in the frequency of variants. As seen in Figure 4, under pure neutrality the probability of adopting a variant

is directly proportional to its current frequency (dashed line). However, when positive content bias is introduced, adoption probability increases proportionally to both the bias strength and variant's frequency. In addition, population size (N) and bias strength (b) interact in the following way: in small populations, drift can overwhelm weaker biases, whereas in larger populations subtle biases can strongly shape variant frequencies.

In archaeological contexts, the quality or attractiveness expressed by the bias strength can correspond to functional properties, aesthetic preferences, social prestige or other trait-specific advantages influencing adoption (Dunnell, 1978; Henrich and Gil-White, 2001; Henrich and Boyd, 2002; Berl *et al.*, 2021). Therefore, explicitly incorporating content bias into cultural transmission models enables archaeologists to better understand how these biases may have shaped cultural assemblages and influences patterns observed in the archaeological record.

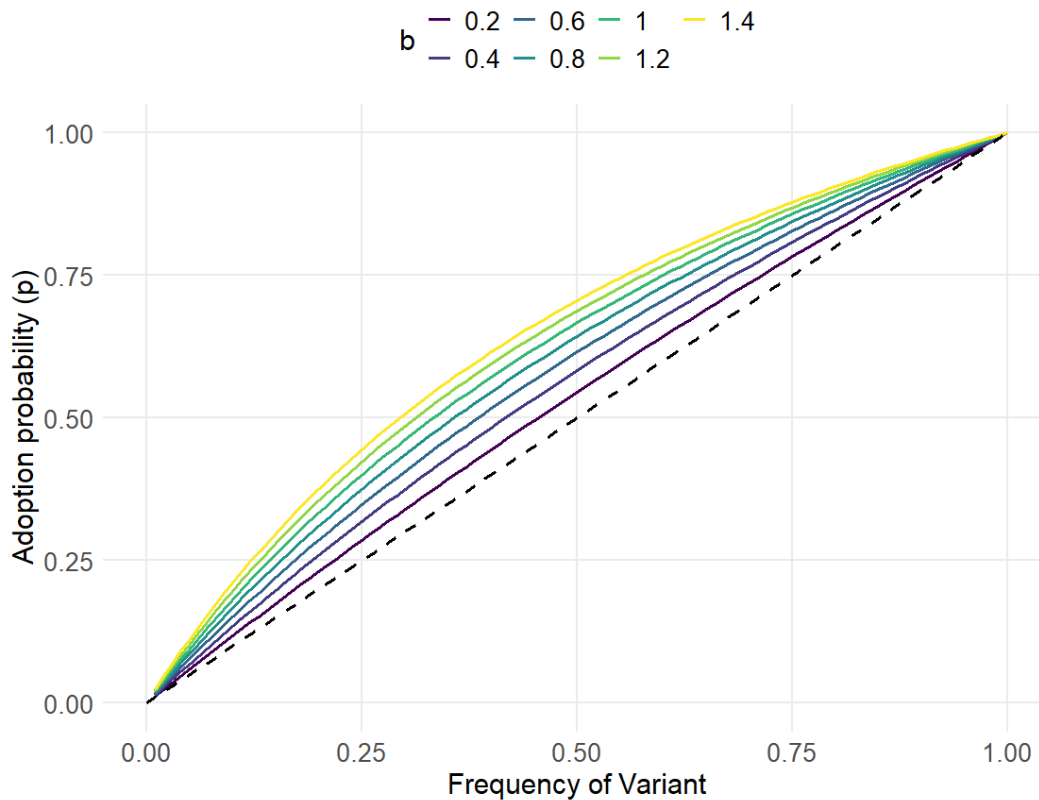


Figure 4: Effect of positive content bias on adoption probability. Solid curves show the probability of adopting variant i as a function of its population frequency (1-100%) under content-bias strengths ranging from 0.2 to 1.4; dashed line indicates $b = 0$. Increasing b systematically increases the adoption probability of more frequent variants, illustrating how stronger bias amplifies direct biased transmission.

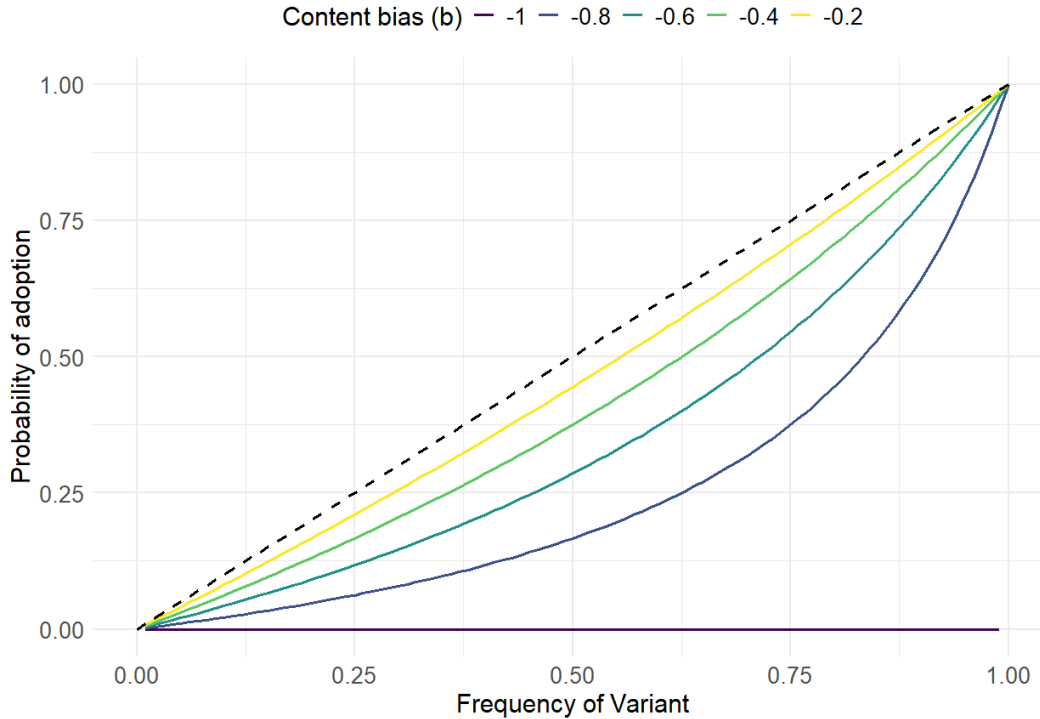


Figure 5: Effect of negative content bias on adoption probability. Solid curves show the probability of adopting variant i as a function of its population frequency (1-100%) under content-bias strengths ranging from -1 to -0.2; dashed line indicates $b = 0$. The plot showcases how when $b = -1$ the “attractiveness” score is 0 ($a = 0$), so the variant goes extinct.

3.2.3. Conformist biased transmission

In the frequency-biased transmission model, variants are adopted preferentially based on their relative frequencies within the population. This model captures conformist bias, a form of frequency-dependent selection wherein individuals disproportionately imitate the most common variants, enhancing their dominance over time (Boyd and Richerson, 1985; Henrich and Boyd, 1998).

Under this transmission scheme, we first run a neutral burn-in phase, allowing variant frequencies to stabilise around equilibrium through random copying and innovation. At equilibrium, we identify the most common variant present at this stage and designate it as the focal variant for conformist bias. Unlike previous models where selection advantages are constant, conformist transmission provides an adaptive advantage that explicitly scales with frequency. Thus, it succeeds to imitates real-life scenarios where more frequent traits are considered more advantageous or socially preferable (Henrich, 2001).

Specifically, at every transmission event following equilibrium, each individual adopts

variants from the previous generation based on frequency-dependent probabilities. The focal variant receives an increase in adoption probability (Fig. 6), weighted by its relative frequency raised to a conformist exponent $(1 + c)$, while other variants maintain neutral constants. Formally, the probability of adopting the focal variant at a given time step is expressed as:

$$P(i) = (1 - \mu) \frac{f_i^{1+c}}{f_i^{1+c} + \sum_{j \neq i} f_j}, \quad (6)$$

where f_i is the frequency of a single i th variant and f_j the frequencies of all the other variants in the population.

This formulation ensures that the bias in favour of the focal variant is dynamically tied to its prevalence in the population. It effectively simulates a scenario in which the focal variant becomes increasingly selected due to its commonness. The parameter c controls the intensity of the conformist bias: when $c = 0$, transmission is strictly neutral; as c grows, the bias strengthens, disproportionately amplifying the success of the most frequent variant. We model conformist bias thus way to isolate the effects of conformity so that FIT can capture them more easily.

In archaeological terms, conformist bias might represent scenarios where widespread traits become popular because of their visibility or perceived social acceptance. Examples include decorative motifs whose prevalence reinforces their cultural desirability independent of intrinsic functional advantages (Kohler et al., 2004; Porčić, 2015). Evaluating SST's performance applied to this model enables us to determine the sensitivity of power in detecting frequency-dependent forces that shape cultural assemblages.

3.3. Archaeological time schemes

one idealised, and another as an alternative generative process

To each simulated model we apply two types of time schemes: *snapshot* and *time averaging*, each representing alternative generative processes of archaeological assemblages. It is worth noting that for each scheme we keep sample size fixed at the population size N , instead of subsampling the population (as done by Feder, Kryazhimskiy and Plotkin (2014) and Crema, Kandler and Shennan (2016)). By doing so we ensure that any change in FIT performance is driven by the temporal structure of the data, rather than any differences in per-sample collection effort or sampling error.

3.3.1. Snapshot

The snapshot scheme simulates assemblages deposited during each event in the transmission chain after equilibrium, reflecting short-term fine-grained depositional events. It essentially emulates an idealised scenario in which archaeologists directly observe each discrete transmission event after

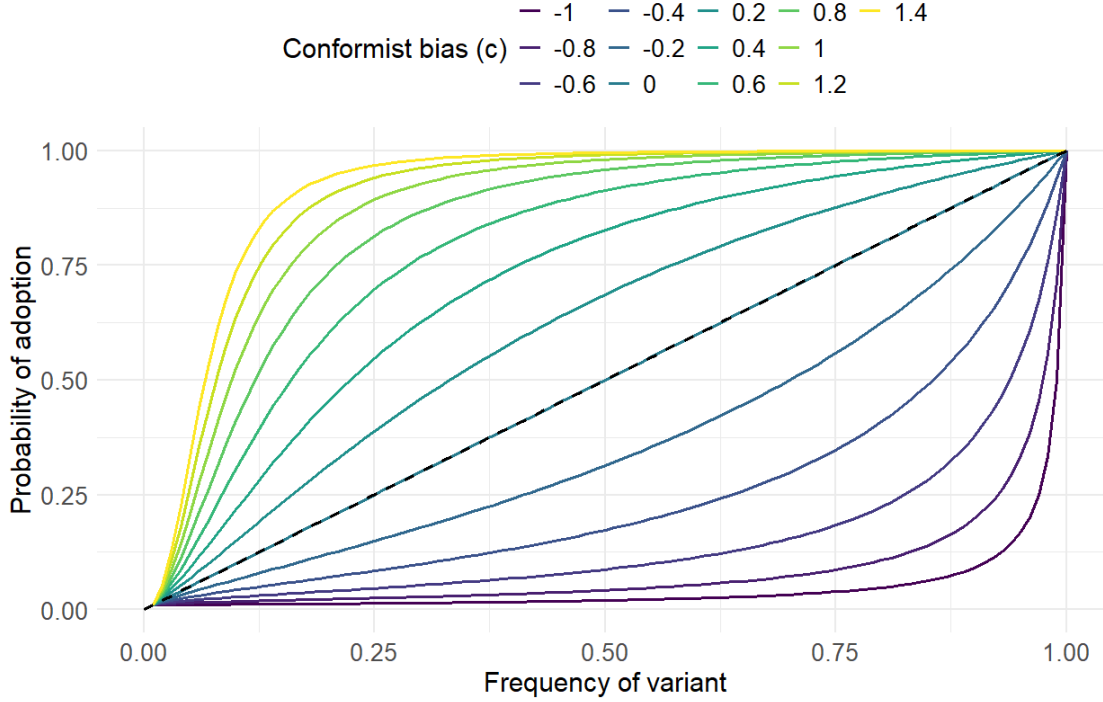


Figure 6: Effect of conformist bias on adoption probability. Solid curves show the probability of adopting variant i as a function of its population frequency (1-100%) under conformist-bias strengths ranging from -1 to 1.4; dashed line indicates $c = 0$. Here we clearly show how frequency becomes a stronger correlated variable for probability adoption than in content biased scenarios. With a bias of $c = 0.4$ a variant at frequency 0.5 has a probability of adoption >0.75 , whereas when $b = 0.4$ its probability is <0.6 .

equilibrium has been reached. Although virtually impossible to come across in the archaeological record, this sampling scheme provides a useful baseline for comparison. By emulating ideal theoretical conditions, we aim to create a reference to which compare other sampling regimes that reflect a more realistic approach.

Within the simulations snapshots are recorded for each discrete time step, which equals parameter t , i.e. equilibrium generations. For every snapshot we record the frequency of all variants (size N) and count the time points (rows) each unique variant appeared in (Fig. 7). This produces a frequency distribution that highlights every single fluctuation within the time series, unlike the smoothed patterns seen in empirical assemblages (Fig. 1). Thus, there is no chronological uncertainty that could produce a confounding effect.

3.3.2. Time averaging

Time averaging is a well-known phenomenon present in the archaeological record. It describes the temporal accumulation of artifacts (often called palimpsests) which have been

irregularly abandoned and deposited but seem to be contemporaneous when excavated (Lyman, 2003; Madsen, 2012; Premo, 2014). By simulating the temporal aggregation of evolutionary dynamics within time series we aim to describe a set of outcomes which are theoretically more akin to empirical distributions. In fact, temporal dynamics described by genetic neutral models have been argued not to coincide with the archaeological reality. As put by Bentley et al., (2004): “Unfortunately, the equilibrium distribution given by Kimura & Crow (1964) is the distribution at a single point in time, whereas archaeologists and anthropologists usually encounter data representing the accumulation of cultural variants over time [. . .]” (p. 1446).

For the snapshot version every row represents a time step, and each column is one individual carrying a variant, thus at each time step we record N observations. However, within the time averaging scheme we aggregate t into time windows of size w , thus each row does not represent the actual time series defined by t , but the averaged time bin (or archaeological layer/stratum). Thus, if $w = 20$ and $t = 1000$, the number of time bins/rows simulated is 50 with a duration of w . Moreover,

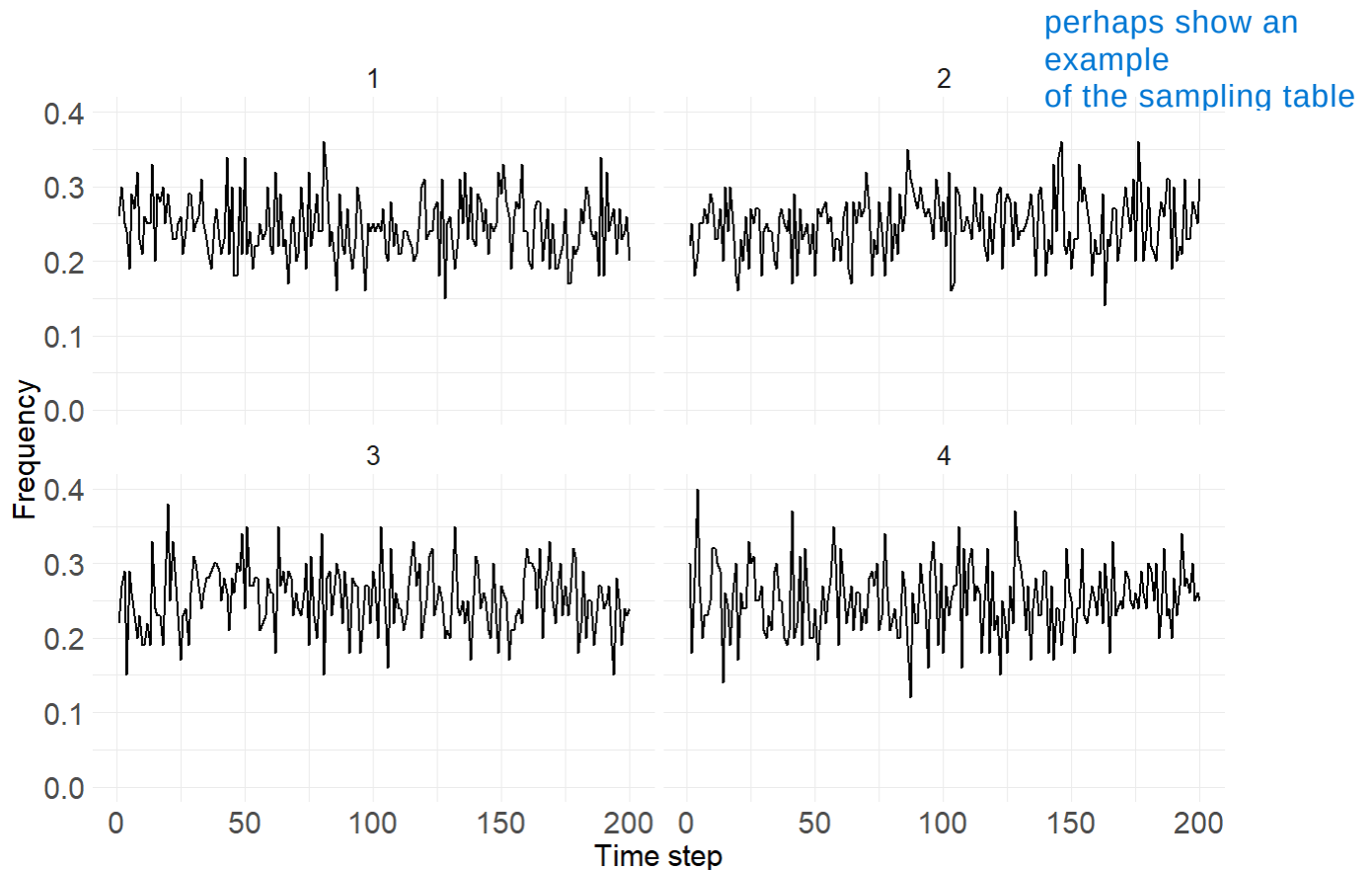


Figure 7: Simulated time series of variant frequencies under unbiased transmission at equilibrium ($N = 100$; $B = 200$). The model assumes a population of four cultural variants transmitted across successive time steps without selection. Frequencies reflect individual-level copying with innovation ($\mu = 0.02$), recorded under snapshot conditions (no time averaging).

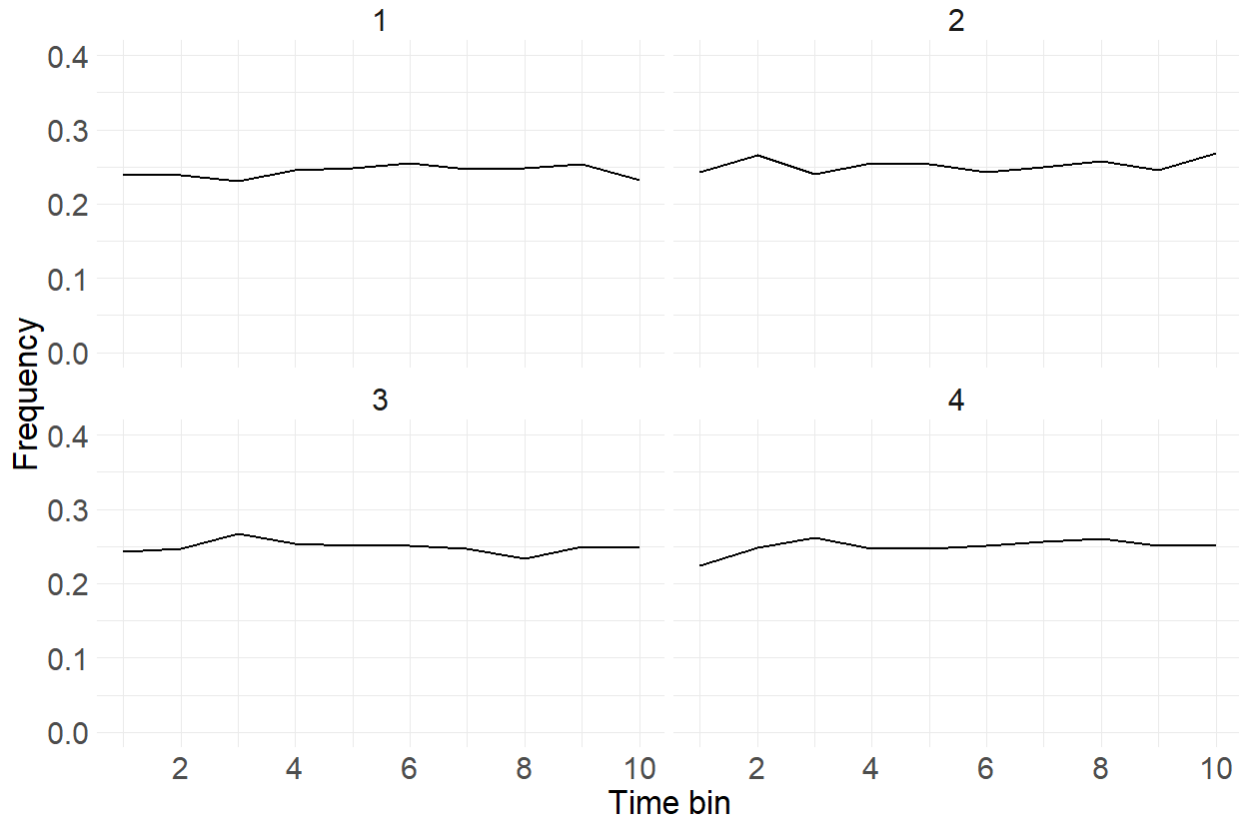


Figure 8: Simulated time series of variant frequencies under neutral (unbiased) transmission with time averaging. As in Figure 7, the population consists of four cultural variants transmitted through unbiased copying and innovation. Here, time averaging aggregates data across fixed time windows, emulating the effects of archaeological assemblage accumulation. The result is a smoothed frequency signal, with reduced resolution of short-term fluctuations.

if our population has N artefacts per generation and we excavate a layer that accumulates over w generations, then up to $N * w$ artefacts are “discarded” into that layer, which translates into $N * w$ observations per row.

This strategy considerably inflates the number of potential variants tested by FIT, a potential factor that could increase the test’s power to detect both neutral and content biased variants (see Chapter 4). In contrast to the snapshot scheme, time averaging smooths out the small-scale short-term transmission events, as they are averaged out by aggregating time steps into a factor w (Fig. 8).

Finally, although real strata vary in thickness and depositional rate, we hold w constant across simulations to isolate its effect. Nonetheless, we sweep over multiple w combinations to explore how time averaging affects SST power, but always keeping the variable constant within each simulation.

3.4. Experimental design

The main objective is to evaluate the performance of the test through a series of controlled simulations entailing cultural transmission models: neutral evolution, when transmission is unbiased; and selection, whether applied to a variant through an “attractiveness” parameter (content bias), or through a replication bonus tied to their frequency at any given time (frequency-based). In addition, we combine each transmission model with a time scheme, thus having a total of six models to test.

The methodological apparatus is comprised by two simulated experiments: Experiment 1 – Neutral Detection and Experiment 2 – Signals of Selection. Both experiments share the same core parameters N , μ , B and t , but the latter accounts for the added parameters b and c —content and conformist biases respectively. To achieve the goal of the analysis, we put forward a set of evaluation metrics tailored for each experiment—Neutral Detection Rate (NDR, proportion of neutrally detected variants) and Signal of Selection Rate (SSR, proportion of biased variants with $p < 0.05$), combined with their respective Type I error (False Positive Rate) and Type II error (False Negative Rate) (Cohen, 1988, 1992). Because FIT cannot run the one sample t -test without three time points (“three-time-point rule”), we report that these metrics represent non-NA results, i.e. those instances where the test successfully ran the t -test and returned a p -value. In fact, we report the proportion of NA variants (%NA), which we interpret as caveats in which FIT could not run due to a variant observed in less than three time points.

For each experiment, we first run one parameter combination for 100 runs to observe non-varying conditions. All parameters are set to the following values: $N = 100$, $\mu = 0.01$, $B = 1000$, $t = 1000$; when applicable $b = 0.15$ and $c = 0.05$. They remain constant throughout the simulations, thus providing a baseline reference run to which we introduce experimental treatments. This is followed by a parameter sweep stage which consists of running the simulation for each parameter combination, illustrated in Tables 1 and 2.

Simulations and FIT application were carried out in R language version 4.5.0 (R Core Team, 2025). Full, version-controlled code and a fixed random-seed policy (`set.seed(1234)` at the start of every pipeline) are publicly available at <https://github.com/Cooperr18/Dissertation-Code>. The following subsections describe Experiment 1 (neutral detection) and Experiment 2 (selection power) in turn, using the common framework outlined above.

3.4.1. Experiment 1

We simulate a population of N size with a μ probability of innovation. Each time step every generation is reshuffled, thus individuals can either imitate an existing variant or adopt a novel one by innovating. After running this process for B generations, the population reaches a stationary

equilibrium, and we start recording transmission events for t time steps. For every generation there is a N number of individuals each carrying a copy of a variant. When the matrix of the population is completed, we store all unique variants throughout the time series and count the time points in which each variant has been present. Variants are then filtered by time points—those of less than three time points are excluded in the analysis, then the FIT is applied.

Once we have ran the one sample t -test we store every variant’s p -value ($\alpha = 0.05$) and flag them as *neutral* (> 0.05), *selection* (≤ 0.05), and *NA*—if FIT could not complete the t -test. We then count NA and non-NA (neutral/selection) variants separately and compute a ratio of both. We repeat this process n_{runs} number of runs and calculate the mean of both outcomes. This results in the Neutral Detection Rate (NDR) and False Positive Rate or Type I Error (FPR) for non-NA variants, and the proportion of NA for the failed-to-detect variants (%NA). An adequate accuracy to accept the null should not account for less than 0.95 replicates, i.e. FPR no greater than 0.05 (Cohen, 1988). In parallel, the share of NA outcomes be negligible, ideally close to 0, to signal that data structure rather than test logic is not limiting performance.

In the snapshot version NDR represents the number of times throughout the complete t time series that FIT flagged a neutral variant. Whereas in the time averaged version NDR is bounded to the number of aggregated bins (t/w), although it is computed with a higher number of observations per time step.

For both time schemes, we run a single baseline pipeline (100 replicates), and multiple parameter sweeps (1000 replicates each)(Table 2). The sweep systematically varies population size N , innovation rate μ , time series length t , and time window size w with the combinations specified in Table 2. We expect (i) larger N and longer t to lower FPR and boost NDR, (ii) higher μ to raise

N	μ	t	w
10	0.01	100	5
50	0.025	200	10
100	0.05	500	20
150	0.075	750	40
200	0.1	1000	50
250	0.125	1500	70
300	0.15	2000	80
350	0.175	2500	100
400	0.2	3000	200

Table 1: Experiment 1 parameter sweep combinations. Note that when a parameter varies the rest stay at baseline levels, specified by the baseline pipeline: $N = 100$, $\mu = 0.01$, $B = 1000$, $t = 1000$ and $w = 20$.

N	μ	t	w	b	c
10	0.01	10	5	0	0.01
50	0.025	50	10	0.05	0.05
100	0.05	100	20	0.1	0.1
150	0.075	150	40	0.15	0.15
200	0.1	200	50	0.2	0.2
250	0.125	250	70	0.25	0.25
300	0.15	300	80	0.3	0.3
350	0.175	350	100	0.35	0.4
400	0.2	400	200	0.4	0.5
500	0.25	—	—	—	0.6
600	0.4	—	—	—	0.7
700	—	—	—	—	0.8
800	—	—	—	—	—
900	—	—	—	—	—
1000	—	—	—	—	—

Table 2: Experiment 2 parameter sweep combinations. Note that when a parameter varies the rest stay at baseline levels, specified by the baseline pipeline: $N = 100$, $\mu = 0.01$, $B = 1000$, $t = 1000$, $w = 10$, $b = 0.15$ and $c = 0.05$.

variant richness and slightly inflate NA counts, and (iii) moderate w values to smooth noise out improving NDR.

3.4.2. Experiment 2

This experiment follows the same pipeline as Experiment 1: a burn-in stage of B generations to reach equilibrium, then running FIT under snapshot and time averaging schemes separately. The key change is that we introduce a *focal variant* with an extra probability of transmission provided by their respective b or c bias. We adopt two different strategies for content and conformist biased transmission.

In the content-biased model, at equilibrium, we add the focal variant whose weight a exceeds 1 (Eq. 5). All other variants stay at $a = 1$. Because we aim to detect selection when it is present, we must add at least one focal variant per run ensuring that it survives three time points or more. For this reason, we inject it with an initial frequency whose probability of extinction after the third time step is vanishingly small. We calculate the minimum initial count k so its chance of extinction in two generations stays below 5 %. We solve

mention that this is based on previously failed experiments which did not detect

$$(1 - k/N)^{2N} < 0.05$$

and seed k carriers at $t = 1$:

$$k = N \left(1 - 0.05^{\frac{1}{2N}} \right)$$

This gives us a $\geq 95\%$ chance that the focal variant appears in all three early time steps, thus minimising potential NA results. We have considered other injection strategies, such as (i) hard-coding the focal variant’s ID, always greater than N , (ii) assigning the focal to the modal variant—highest time points—from the burn-in stage, and (iii) randomly sampling any variant from the burn-in stage that at least scores three time points. Nonetheless, these are based on somewhat arbitrary conditions, and most importantly had greater chances of extinction².

When transmission is conformist-biased, we pick the variant with the highest count at equilibrium (highest initial frequency) and assign it the c bias. We have opted to apply other biases such as pro-novelty or anti-conformist bias. We reason that conformist bias provides a clearly defined scenario of positive frequency-dependent selection, because it ensures that the focal variant gains an explicit advantage. Conversely, pro-novelty and anti-conformist biases would provide selective advantages to variants with lower initial frequencies. However, as we reason in Chapter 4, such changes would only be detected by FIT if they escalated moderately. How moderate is a matter that is yet to explore, so we leave it for further investigation. This could be mentioned above

When the population is completed at $t = 1000$, we follow the same procedure, but this time we subset the trajectories of the focal variants across runs. We sum neutral ($p > 0.05$) and selection ($p \leq 0.05$) flags along with NA variants, and compute the ratio of selection per run (Signal of Selection Rate—SSR) and the False Negative Ratio (FNR) or Type II Error (adding up to 1), both excluding NA variants.

As in Experiment 1, we conduct the parameter sweeps listed in Table 2. Each unique set of values is held constant throughout its simulation batch, and each combination is replicated $n_{\text{runs}} = 1000$ times to stabilise our estimates. We anticipate that in this experiment, (i) SSR will follow an S-shaped power curve for weak and strong signals at both ends—similarly to the s power curve outlined by Feder, Kryazhimskiy and Plotkin (2014) (Fig. 9), (ii) larger N will improve power and reduce both false positives and NAs, and (iii) intermediate time averaging windows (w) will strike the best balance—smoothing out noise enough to boost SSR without erasing signals of selection.

For conformist-biased transmission we apply a parameter sweep through bias strength (c) and population size (N). We expect (i) stronger c to yield higher SSR, in a similar fashion to b ,

²Other alternatives to focal variant injection were discussed and discarded throughout the experiment design process. To provide full transparency, see “Report of daily progress” for a more detailed argumentation, available at <https://github.com/Cooperr18/Dissertation-Code>.

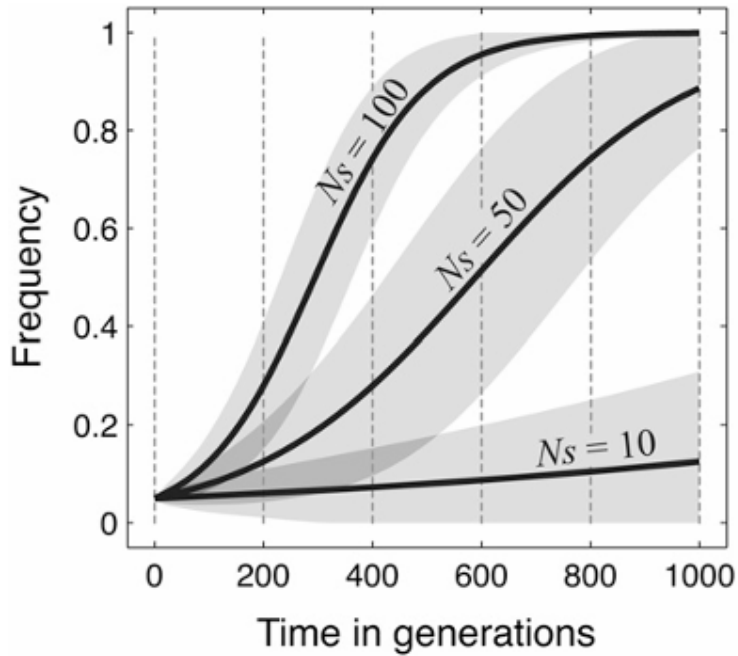


Figure 9: Power curve of the coefficient of selection, extracted from Feder et al. (2014, fig. 3). The diagram describes the power of coefficients $s = 0.001, 0.005, 0.01$ throughout 1000 generations. As pointed out by the authors, when selection is weak ($Ns = 10$) the frequency changes are so nuanced that FIT fails to detect selection. On the other hand, when selection is too strong ($Ns = 100$) fixation events make the frequency series uninformative at the end of the time series, leading to a loss of power. Ideally, mid-levels of s at a middle sampling interval should result in the best power.

where the test loses power at both ends of the curve. Moreover, (ii) SSR is expected to be lower and variable with small populations, reflecting increased effects of drift and noise. Finally, (iii) under time averaging we expect SSR to decrease overall due to data smoothing, particularly at weaker bias intensities.

Chapter 4. Results and Discussion

In this combined section we integrate empirical findings from two experiments—neutrality detection (Experiment 1) and content & conformist bias detection separately (Experiment 2)—with their interpretation to provide a cohesive narrative. Within each experiment, we first lay down a baseline simulation with fixed parameters, followed by single parameter sweeps, and finally multiple parameter interactions. Experiment 1 describes the SST’s capacity to detect unbiased transmission and variation, thus laying the foundations upon which Experiment 2 departs to assess power to detect selection.

4.1. Experiment #1 – Unbiased Transmission

4.1.1. Results

When no selection is present, does the test correctly identify variants as neutral between snapshot and time averaged schemes, i.e. accepting the null hypothesis? Under default parameters ($N = 100$, $\mu = 0.02$, $B = 1000$, $t = 1000$, $\alpha = 0.05$, and 100 runs) when all time steps are sampled—snapshot version—neutrality is widely detected across runs. The Neutral Detection Rate (NDR) at equilibrium conditions across runs is 0.998, and the False Positive Rate (FPR, or Type I error) equals 0.001 (Fig. 10), thus surpassing the expected threshold of accuracy. However, the hit-rate only accounts for non-NA variants that scored at least three time points (“three-time-point rule”), meaning that those variants that could not be analysed by the test (< 3 time points) were not included. In fact, if we compute the hit-rate counting all unique variants, NDR goes down to an average of 0.826 (Fig. 11), and the proportion of non-detected variants remains similar (17.6 % and 17.2 % respectively).

At the implementation level, the FIT struggles to compute the t -test for variants that remain near around three-time points. This issue is consistent across runs (Fig. 12), with %NA values oscillating around 17.2 %, scoring 13 % the lowest and 21 % the highest. Therefore, under pure drift, and with information about single transmission and depositional events, short-lived variants are not detected due to their small representation throughout the time series. Moreover, because the model at use assumes that variants cannot reappear, when drift drives a short-lived variant to extinction its chances of showing up again are effectively 0.

These result contrast with what we report for the time averaged version. We quantified the NDR under pure drift, aggregating 1000 time steps (t) into time bins (w) of size 20—i.e. 50 effective generations (Figure 8 depicts the smoothing effect of time averaging). Under the same parameters as the snapshot baseline simulation, and at equilibrium conditions, the NDR is 0.996 and the FPR is 0.004, across 100 runs. An insightful finding is that time averaging significantly drops the proportion of non-detected variants to 0. Therefore, the hit-rate effectively comprises all unique variants throughout the time series and across runs.

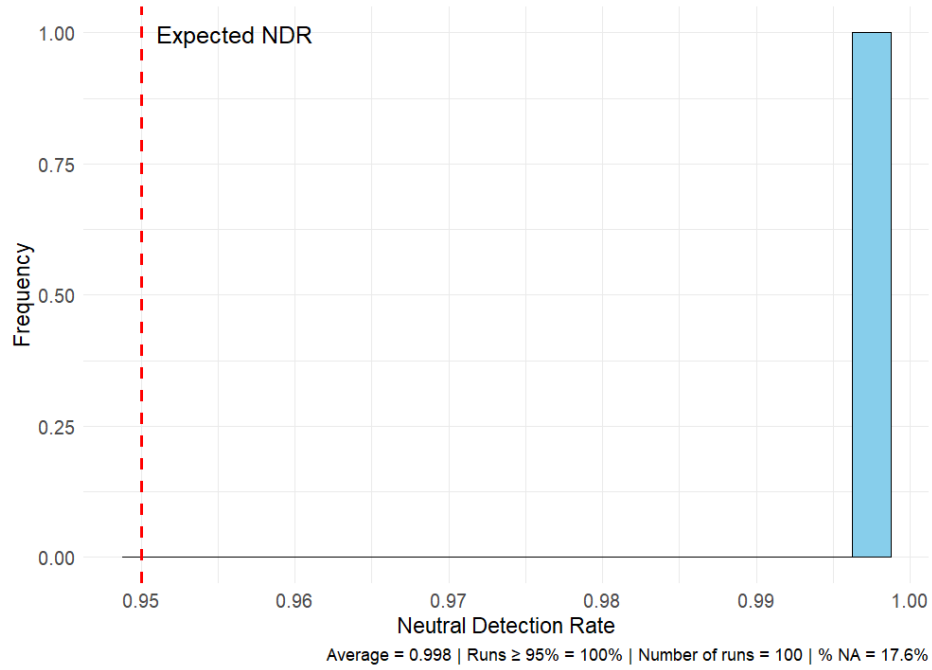


Figure 10: Distribution of Neutral Detection Rate (NDR) under snapshot conditions for variants non-NA cases. The red dashed line marks the expected detection threshold (0.95). Across 100 runs, all simulations exceeded this threshold (average NDR = 0.998). Parameters: $N = 100$, $\mu = 0.01$, $B = 1000$ and $t = 1000$.

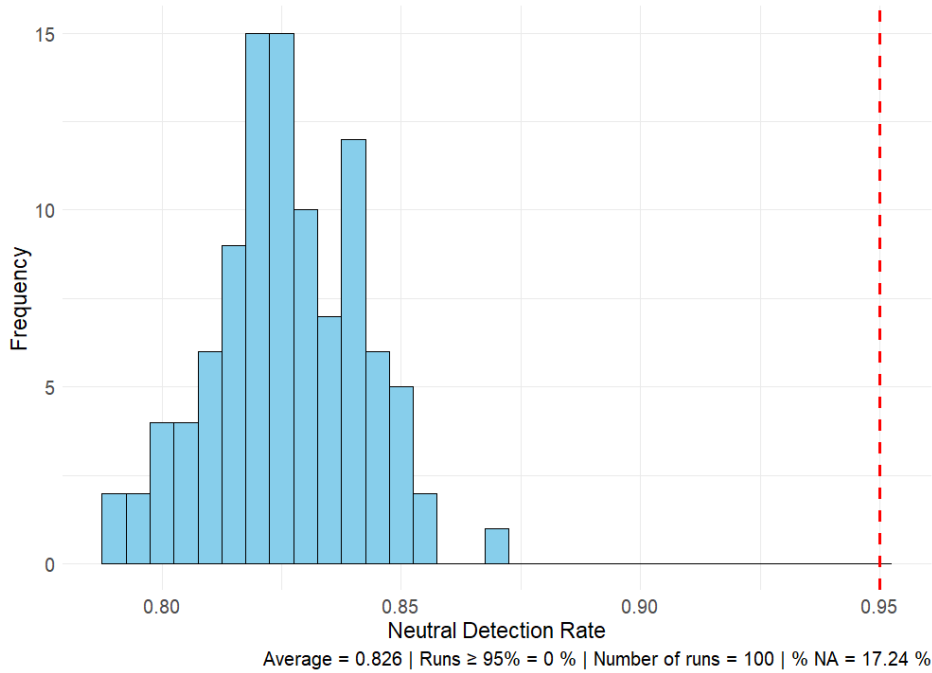


Figure 11: Distribution of Neutral Detection Rate (NDR) under snapshot conditions including all variants, with undetected cases counted. The average NDR drops to 0.826, with no simulation runs reaching the threshold. Same parameters as before.

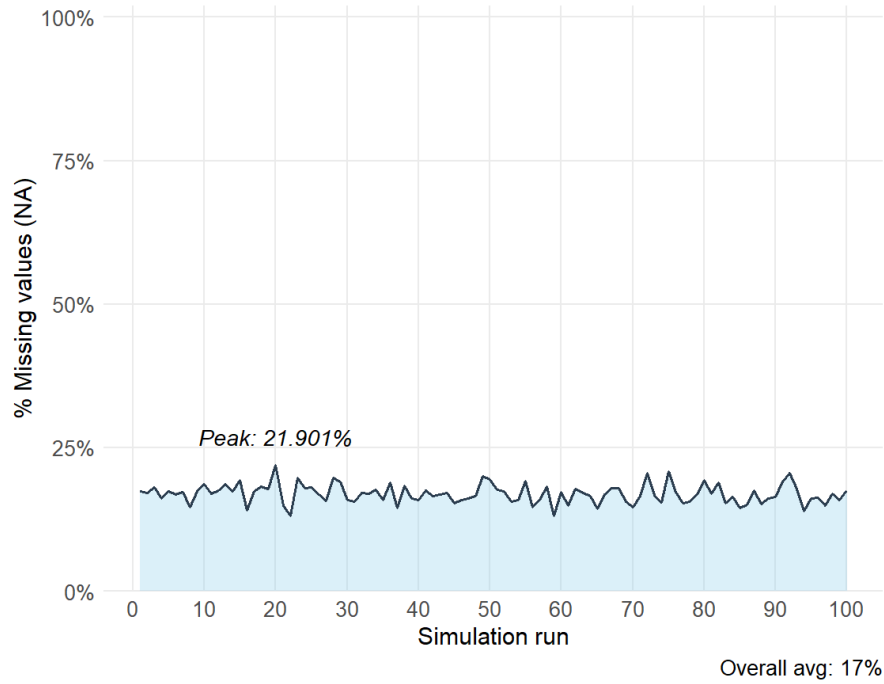


Figure 12: Percentage of missing values (NA) per simulation run under snapshot conditions. The highest observed NA rate was 21.0%, with an overall average of 17.2% across 100 runs.

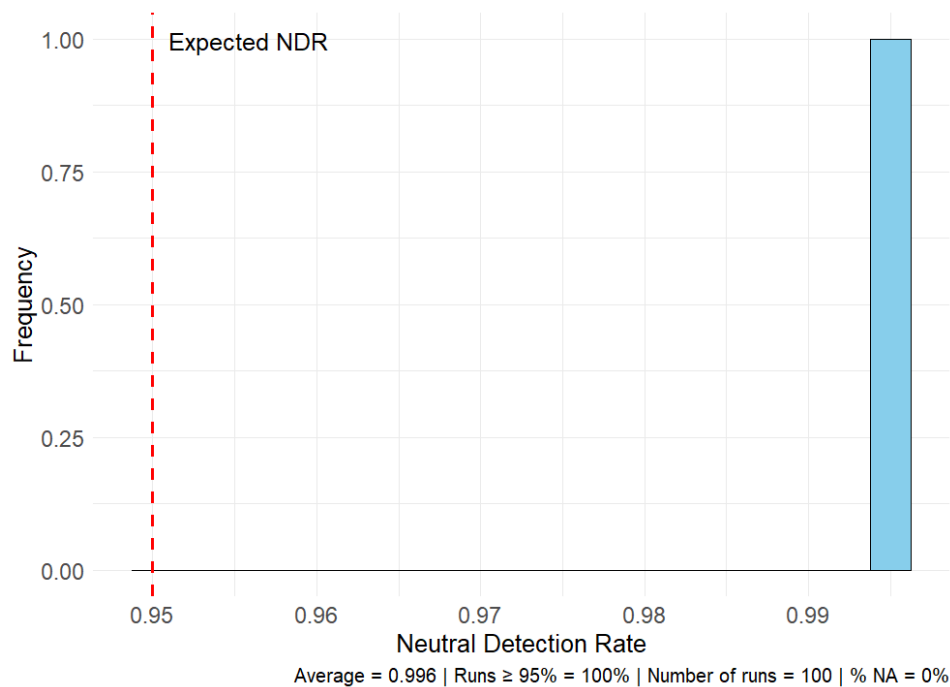


Figure 13: Neutral Detection Rate (NDR) distribution under time-averaged conditions, excluding NA cases. All 100 runs exceeded the 0.95 threshold (dashed line), with an average NDR of 0.996 and no missing values (NA = 0%).

Given the contrasting findings between snapshot and time averaging schemes under default parameters, it is crucial to explore how robust this pattern remains across varying conditions. We performed three univariate sweeps to evaluate how innovation rate (μ), population size (N) and total timesteps (t) influence the Neutral Detection Rate and the proportion of variants excluded by the three-time-point rule (%NA) when all time steps are sampled (snapshot). Summary statistics are reported in Table 3 and illustrated in Figures 14 – 16.

- Innovation rate (μ). As μ increases from 0.01 to 0.2, the non-NA NDR exhibits a slight but consistent upward trend, rising from 0.9978 to 0.999 (Fig. 14). In contrast, the proportion of non-assessable variants increase markedly, climbing from 17.0 % at $\mu = 0.01$ to 25.5 % at $\mu = 0.20$. This pattern reflects the fact that higher innovation rates produce more novel, short-lived variants, many of which fail to meet the minimum conditions of the FIT and therefore are excluded from the t -test.
- Population size (N). Varying between 10 and 400 individuals reveals an inverse relationship

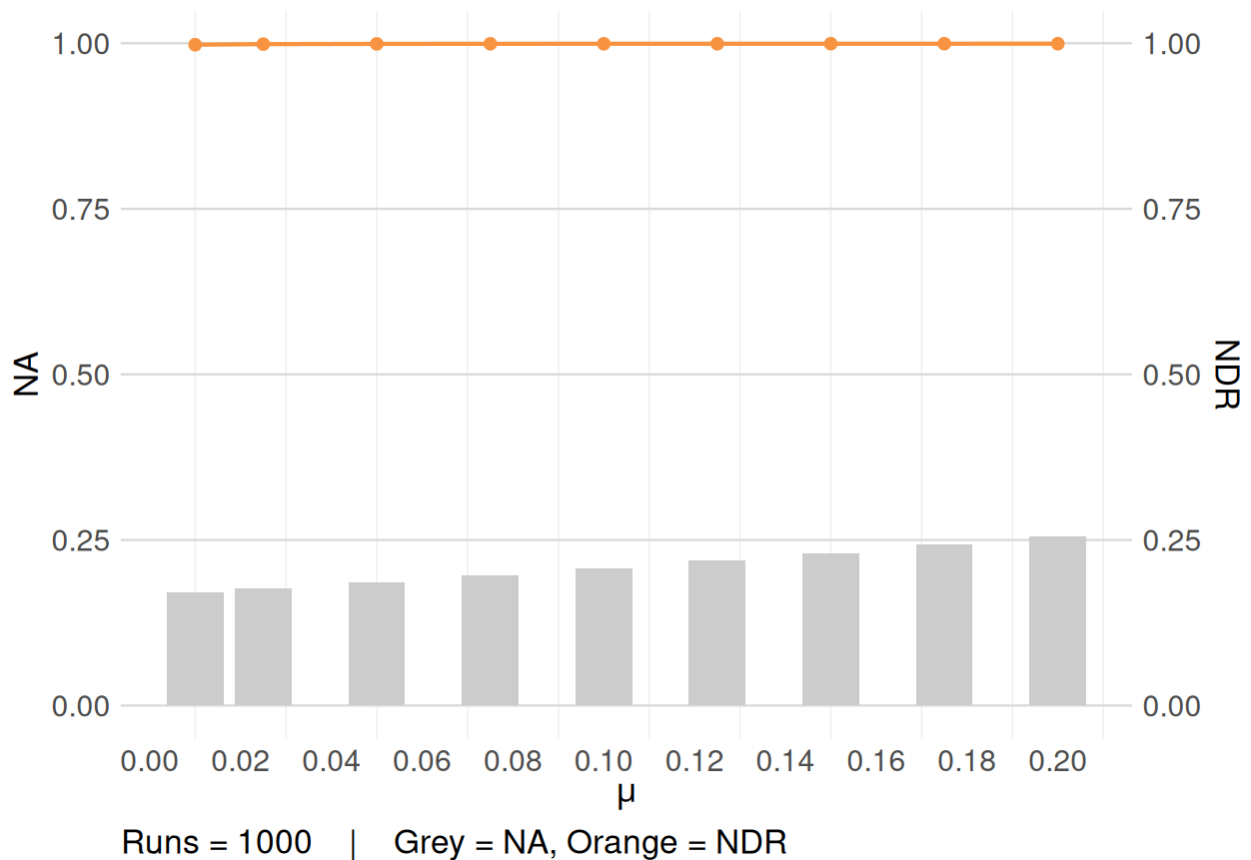


Figure 14: Effect of innovation rate (μ) on Neutral Detection Rate (NDR) and NA proportion under snapshot conditions. NDR slightly increases with μ , reaching a plateau near 0.9993. Higher μ also leads to more missing values, rising from 17% to ~25%.

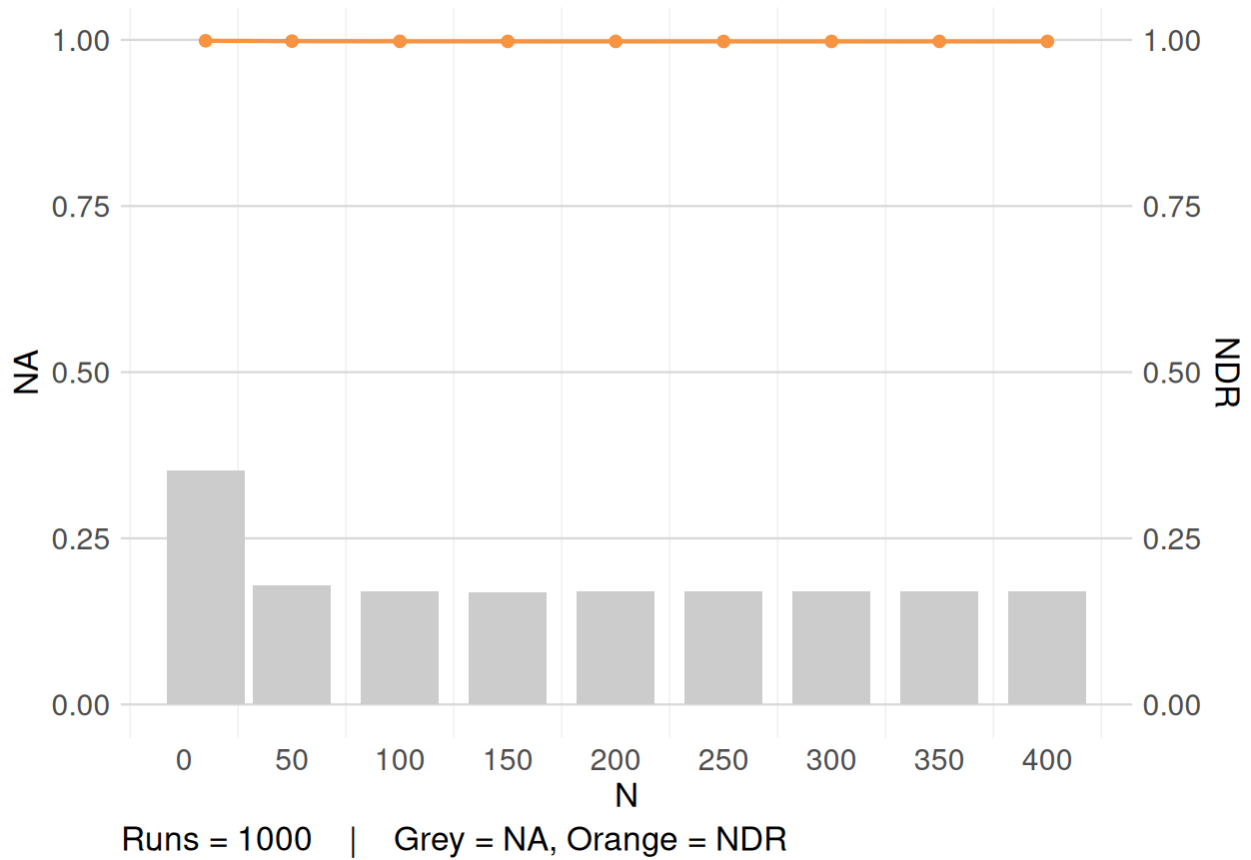


Figure 15: Effect of population size (N) on NDR and NA proportion. NDR slightly decreases with larger N but remains above 0.997. NA drops sharply from 35% to ~17% as N increases.

with NDR: the rate falls modestly from 0.9986 at $N = 10$ to 0.9777 (Fig. 15). By contrast, %NA decreases sharply between $N = 10$ and $N = 50$ —dropping from 35.2 % to 17.0 %—and then stabilises around 17 % for larger populations. The elevated %NA at small N can be attributed to increased stochastic loss of low-frequency variants, which seldom survive long enough to be tested.

- Time series length (t). Extending the number of time steps from 100 to 3,000 produces a steady increase in NDR, from 0.9833 to 0.9989, while %NA remains essentially constant at ~17 % (Fig. 16). Longer time series allow more variants to accrue to the required three observations, thereby both reducing non-detectable cases and modestly improving overall neutral detection.

To further explore the variability of patterns of neutral detection, we also conducted three single parameter sweeps under the time averaged sampling scheme. We explore the same parameters as the snapshot scheme, but adding bin width (w)—i.e. the aggregation of transmission and depositional events (analogy of archaeological strata). Similarly to the snapshot scheme, all NDR

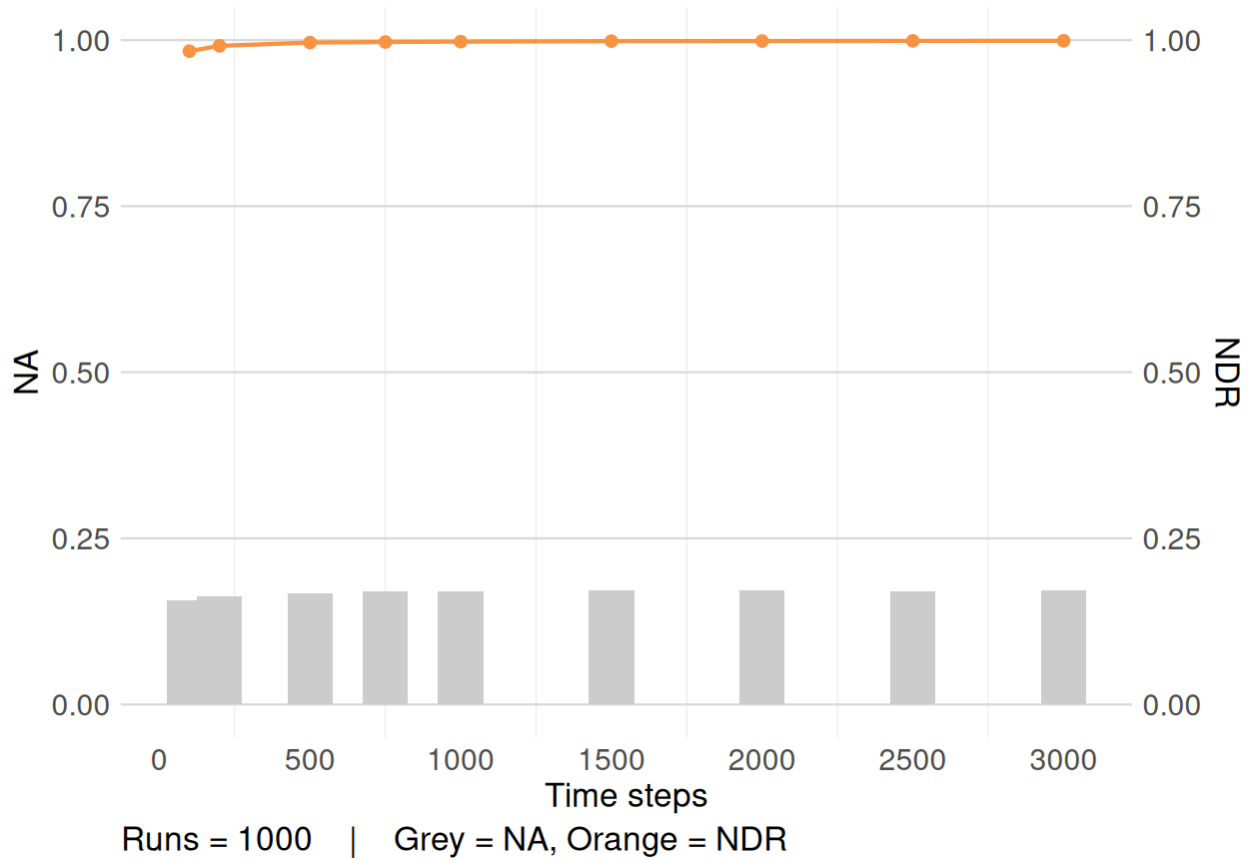


Figure 16: Effect of time series length on NDR and NA proportion. NDR improves with more time steps, plateauing above 0.998. NA proportion stabilizes around 17% after 500 time steps.

values reported correspond to non-NA variants analysed by the FIT. Summary statistics are given in Table 4 and plotted in Figures 17 – 20.

- Innovation rate (μ). Under time averaging, NDR increases monotonically from 0.9931 at $\mu = 0.01$ to 0.9994 at $\mu = 0.20$ (Fig. 17). The proportion of non-assessable variants (%NA) remains essentially zero throughout ($\leq 0.001\%$), indicating that aggregating 1,000 time steps into bins of 20 consistently yields at least three observations per variant. The upward trend in NDR arises because larger μ still generates more turnover—and hence more detectable frequency increments—while the binning inflates the number of variants allowing short-lived ones to persist.
- Population size (N). When N ranges from 10 to 400, NDR shows a slight decline from 0.9962 at $N = 10$ to 0.9932 at $N = 100$, then stabilises around 0.9335 for larger N (Fig. 18). Notably, %NA is 28.25 % at $N = 10$ —reflecting that very small populations yield few surviving three binned variants—but drops to zero by $N = 50$ and remains negligible thereafter. This pattern reflects the interplay of drift and time averaging: at very low population sizes, drift-induced

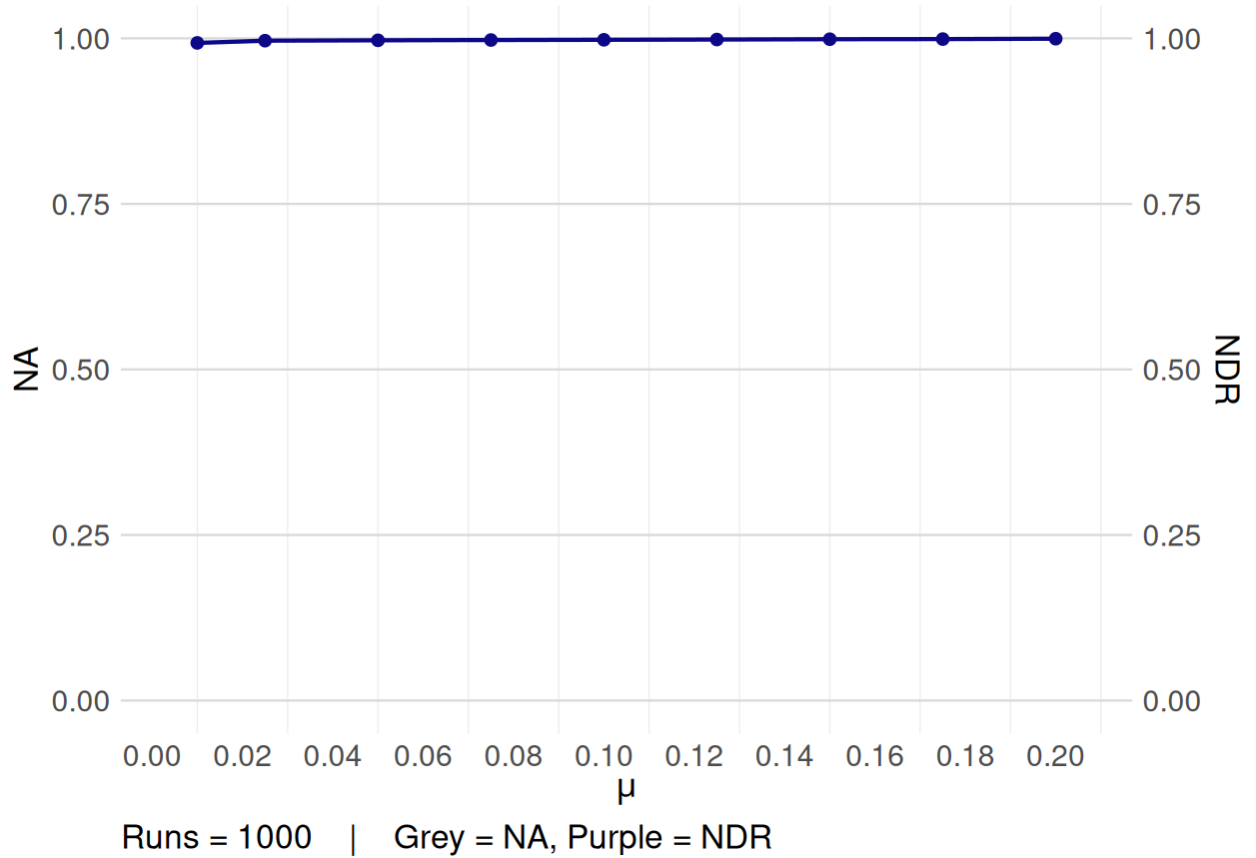


Figure 17: Effect of innovation rate (μ) under time averaging. NDR increases with μ , reaching 0.9994. Missing values are virtually absent across all runs.

extinction still overshadows the effects of time averaging.

- Time series length (t). Extending t from 100 to 3,000 slightly increases NDR from 0.9531 to 0.9968, while %NA remains at or below 0.1% across all series lengths (Fig. 19). Longer series produce more aggregated observations per variant, thereby reducing the few remaining under-sampled cases and enhancing overall neutral detection.
- Time window size (w). Varying bin size w between 10 and 200 ($t = 1,000$ constantly) yields an NDR that oscillates between 0.9923 ($w = 70$) and 0.9961 ($w = 200$) (Fig. 20). %NA is effectively 0 for $w \geq 20$, and at $w = 10$ only 0.02 % of variants fail the three-time-point rule. The dip in NDR at intermediate w reflects the trade-off between smoothing (which reduces variance in frequency increments) and temporal resolution (fewer bins mean larger aggregation, which can mask small neutral fluctuations).

Unbiased transmission revealed a marked contrast between snapshot and time averaged schemes. Under the former, the SST achieved near-perfect NDR (0.998) when only non-NA variants

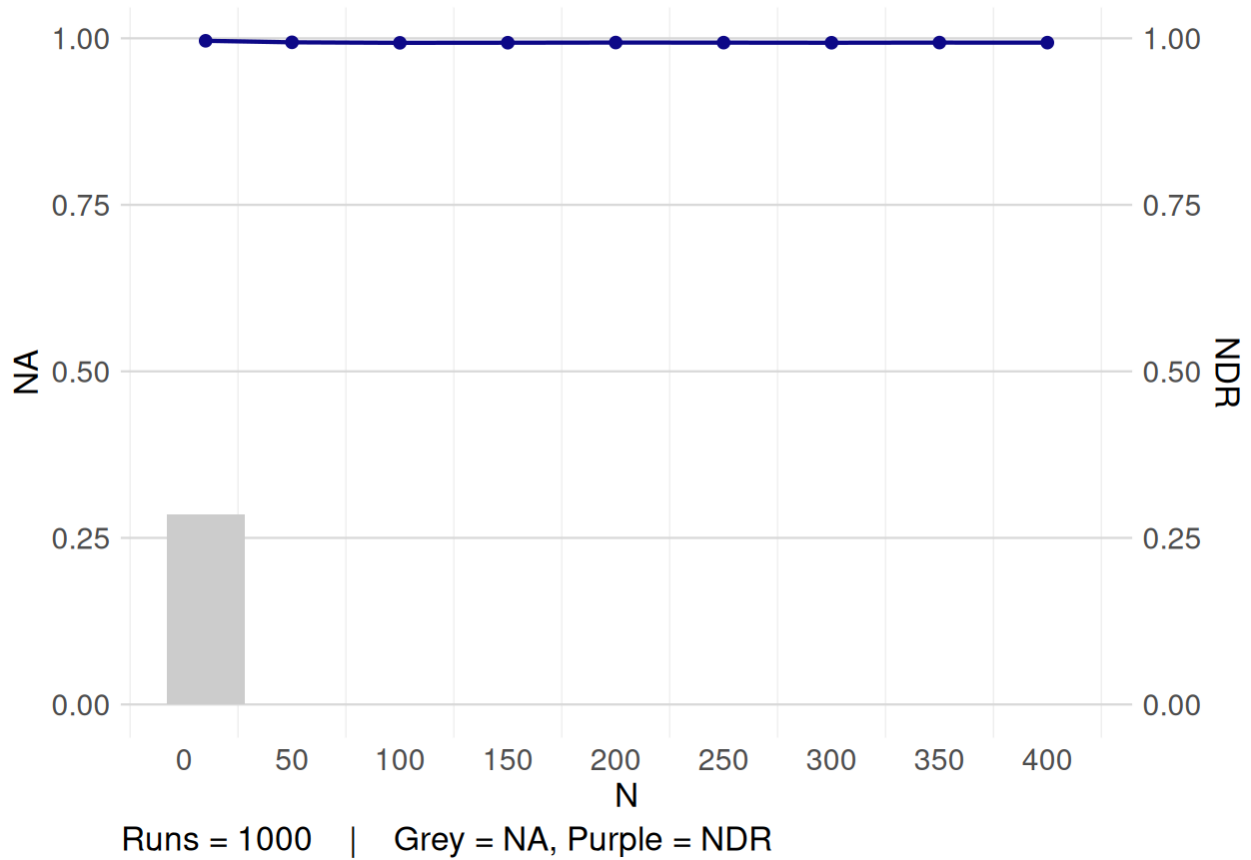


Figure 18: Effect of population size (N) under time averaging. NDR slightly declines with increasing N but stays above 0.993. NA values drop to near zero beyond $N = 50$.

were considered, but ~17% of variants were consistently excluded due to insufficient time points. Higher innovation rates and smaller populations exacerbated this issue by generating more short-lived variants. In contrast, the time averaged model resolved this limitation, yielding consistently high NDR values (>0.99) across sweeps and virtually eliminating NA cases. These results indicate that time averaging enhances detection coverage, making it a powerful tool for identifying neutrality in archaeological data.

4.1.2. Discussion

The central question that drives this first experiment is a prerequisite for any robust application of selection tests in archaeology: How reliable is the SST when there is no selection to detect? Before we can confidently identify signals of selection in the archaeological record, we must first understand the test's baseline performance and its propensity for error under null conditions.

This question directly confronts the methodological struggles outlined in foundational studies (Dunnell, 1978; Neiman, 1995; Shennan and Wilkinson, 2001). Seminal studies like the Merzbach

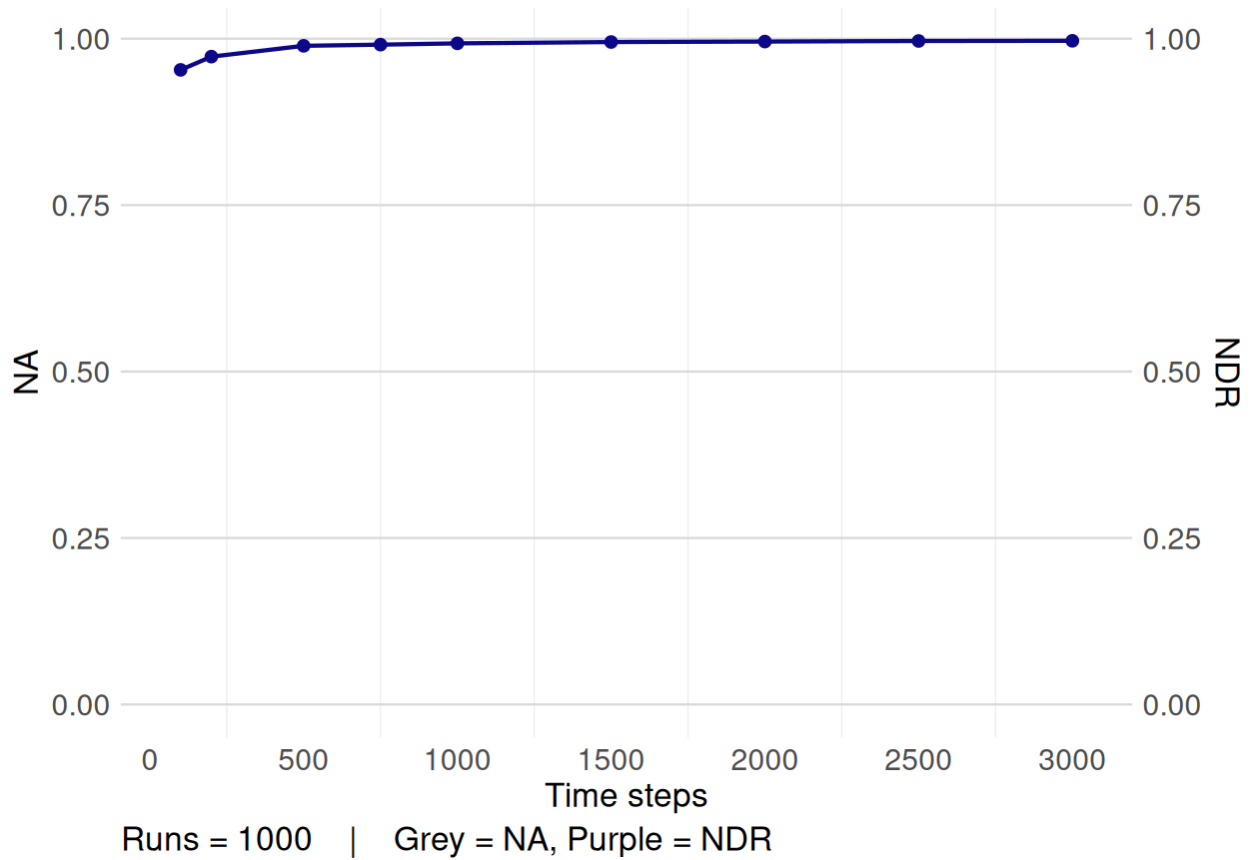


Figure 19: Effect of time series length under time averaging. NDR improves with longer series, reaching 0.9968 at 3,000 steps. NA values remain negligible throughout.

Valley ceramics and Hittite bowls have yielded ambiguous and often contradictory results, largely because the intrinsic error rates and analytical biases of the tests were poorly understood (Shennan and Wilkinson, 2001; Steele, Glatz and Kandler, 2010). Our aim here is thus to move beyond simply applying the Signal Selection Test (SST) and instead to deconstruct its performance under the null hypothesis, thereby establishing a baseline of reliability.

The results unveil a critical insight: SST’s performance is defined by a fundamental dichotomy between accuracy and testability, a division contingent to the time structure of the models. When applied to a high-resolution *snapshot* scheme, the test provides exceptional accuracy. The observed Neutral Detection Rate (NDR) stays consistently over the 0.95 threshold throughout the simulations, which suggests that when the test does flag a selection, it is highly unlikely to be a Type I error. However, this impressive accuracy is achieved at a significant cost. The hidden mechanism for this is what we referred above to as the “three-time-point” rule. It systematically excluded nearly 17 % of variants because of their short lifespan, and we will not know whether they underwent selection or not.

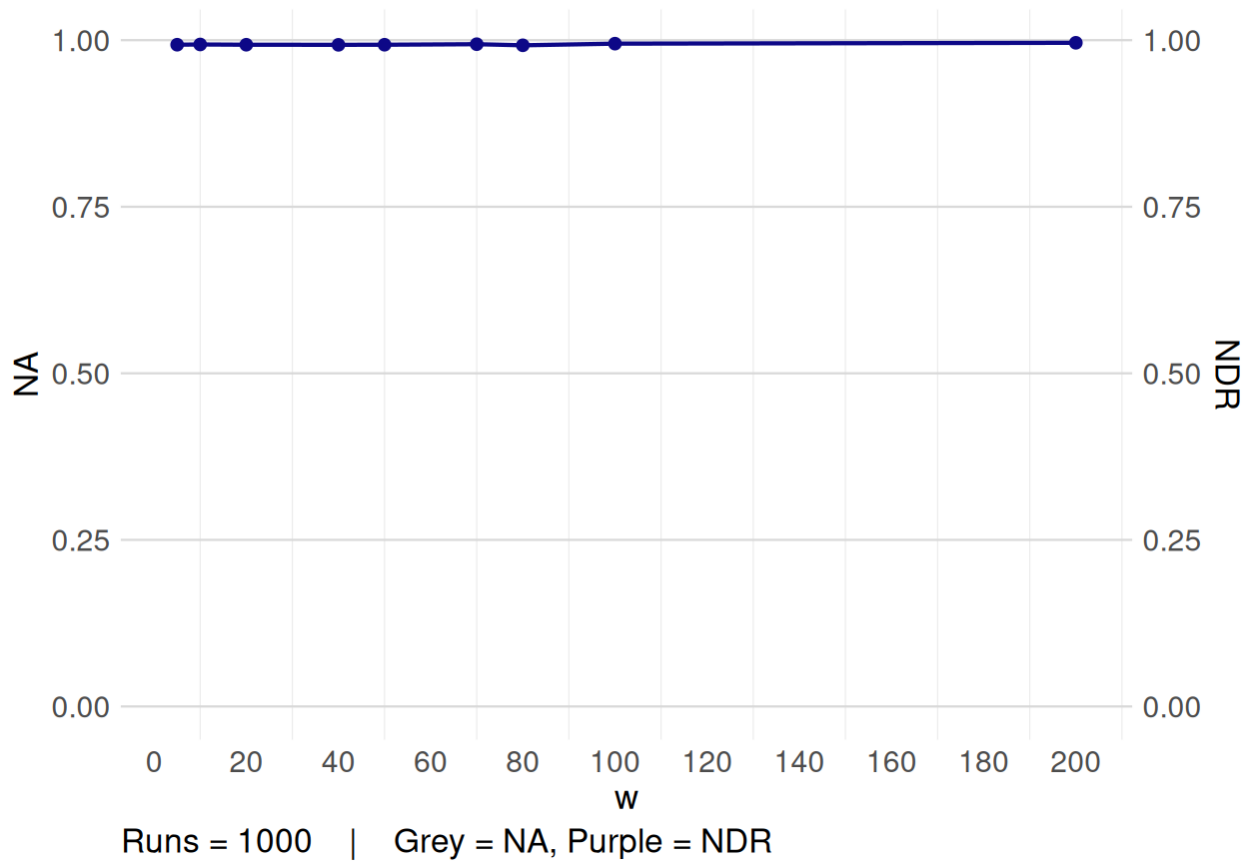


Figure 20: Effect of time window size on NDR under time averaging. NDR increases slightly with larger windows, reaching 0.9961 at 200 steps. NA values remain near zero across all runs.

As revealed in the results, these short-lived variants are disproportionately filtered out, creating a critical bias. These are precisely the kinds of cultural phenomena—failed innovations, copying errors, production biased variants—that were underestimated in the beginning but have now become central to understanding the variability of archaeological assemblages (Eerkens and Lipo, 2005; Kempe, Lycett and Mesoudi, 2012; Eerkens and de Voogt, 2022; Porčić *et al.*, 2024). Excluding them from the analysis could lead to impactful but undetected biases. For example, as theoretically predicted by classic models of cultural drift, in smaller populations the turnover of variants is inherently higher (Wright, 1931; Kimura, 1983; Cavalli-Sforza and Feldman, 1981; Boyd and Richerson, 1985), thus increasing the number of short-lived variants which are rendered analytically invisible by the SST’s snapshot application. Additionally, even if a variant survives three generations, when drift is strong its chances of survival remain low; in these instances, sampling noise blocks FIT of completing the statistical analysis.

It is in this context that time averaging emerges not just as an analytical choice, but as an unexpected benefit. The process fundamentally contradicts the assumption that observed variant

frequencies in assemblages are a close proxy for their true proportions in a past population, a point recently critiqued by Premo (2014) and Crema, Bortolini, and Lake (2024). Yet, our simulations show its practical applicability: by aggregating short-lived variants into larger temporal bins (w), and artificially inflating the number of individuals per bin, time averaging ensures they contribute to a significant frequency count. This explains the near-total elimination of non-analysable variants, and the close-to-unity NDR values across sweeps. What acted as a handicap in the snapshot version is no longer a problem since all variants comfortably surpass the three-time-point threshold.

In essence, this experiment demonstrated that the SST is robust against Type I error. The primary challenge is not a propensity for false positives but a fundamental problem of analytical nature, guided by the interplay between variant lifespan and temporal resolution—a ubiquitous issue in biased transmission models. The snapshot scheme offers a clear but narrow view, while time averaging provides a more comprehensive and panoramic perspective, albeit on a somewhat distorted representation of the past (Madsen, 2012; Premo, 2014; Porčić, 2015; Crema, Kandler and Shennan, 2016). Ultimately, the researcher faces a trade-off: sacrificing fine-grained information to gain more stable and accurate inferences.

4.2. Experiment #2 – Biased Transmission

4.2.1. Results

When one the probability of selection of one variant is increased by content or frequency bias, does the test correctly flag the frequency increase as a selection event? In this experiment we test against the null, i.e. neutrality. To achieve it, the primary metric is the Signal of Selection Rate (SSR), which measures the proportion of simulation runs where the focal variant is correctly detected as selected ($p < 0.05$). This is complemented by the False Negative Rate (FNR), or Type II error, representing cases where a selected variant is misclassified as neutral. Moreover, we report the proportion of non-analysed focal variants (%NA). These instances find an explanation through swift fixation or extinction, a matter we discuss below. We first examine a content-biased model, followed by an analysis of a conformist-biased model, with each model tested under both snapshot and time-averaged time schemes.

4.2.1.1. Content-Biased Transmission

Under the baseline snapshot parameters ($N = 100$, $\mu = 0.01$, $B = 1000$, $t = 100$, $b = 0.15$, $\alpha = 0.05$, and 100 runs), the test demonstrates limited power. The SSR for the focal variant is 0.1818, with a corresponding FNR of 0.8182. The injection strategy for the focal variant proved effective, as the proportion of non-analysable runs (%NA) was 12 %, indicating the variant survived almost

every single run for the required three time points. However, we had to shorten the length of the time series (t) to a 10% from Experiment 1 (1,000 to 100) to guarantee some detection. As we report when sweeping over t , when time series become longer the test loses accuracy, hence the shortening.

This baseline performance is slightly enhanced when time averaging is at play. When keeping the same parameters, but when we aggregate 100 time steps into bins of size 10 (w), SSR increases to 0.457 (FNR = 0.543). However, the proportion of unanalysed variants spikes significantly compared to the snapshot version, up to 54 %. Although achieving higher power, the effective number of variants analysed is smaller, even though the number of “artefacts” increases due to the sampling regime of the model ($N * w$).

Given this modest baseline performance, we conducted four single sweeps, systematically varying the innovation rate (μ), population size (N), bias strength (b), time series length (t), and time window width (w) for the time averaging version, each ran 1,000 iterations³. The results for the snapshot scheme are as follows, and summary statistics are provided in Tables 5 and 6, and illustrated in Figures 21 – 24:

- Innovation rate (μ). Varying between 0.01 and 0.20 yields an overall modest increase in %NA, but SSR decreases impactfully from 0.1075 at $\mu = 0.01$ to 0 at $\mu = 0.1$ onwards (Fig. 21). This finding suggests innovation rate obscures content bias signals under snapshot conditions, and overall, its capacity of testing the focal variant.
- Population size (N). Larger populations find the highest power measures. SSR spikes from 0.12 at $N = 100$ to 0.4095 at $N = 150$, peaking 0.5105 at $N = 300$ and plateauing around ~0.45 for the rest of the sweep (Fig. 22). Following the same trend, %NA decreases with N until it is stabilised around ~11 %. Although not surpassing the 0.95 threshold, larger populations reduce the effect of drift, enhancing detectability and power.
- Content bias strength (b). Increasing bias strength showed no positive correlation with SSR. At lower values, near neutrality, the SSR increases moderately to 0.1481—FNR at 0.8518 (Fig. 23). As b increased, the SSR showed a pronounced dip to 0, oppositely to %NA, which reaches ~80 %. This pattern indicates a low sensitivity of the SST to content-bias changes, and an overall modest power, highlighting its ineffectiveness under clear selection pressures.
- Time series length (t). Longer time series dip SSR values from 0.4027 at $t = 50$ to 0 for most of the sweep (Fig. 24). Moreover, %NA increases with time series, indicating that shorter diachronies are more aligned with FIT’s requisites. This finding underscores the necessity for adequate temporal resolution to accurately detect content bias.

³We also conducted a univariate sweep of number of runs for both schemes using the baseline parameters. Both data and scripts can be accessed at <https://github.com/Cooperr18/Dissertation-Code>.

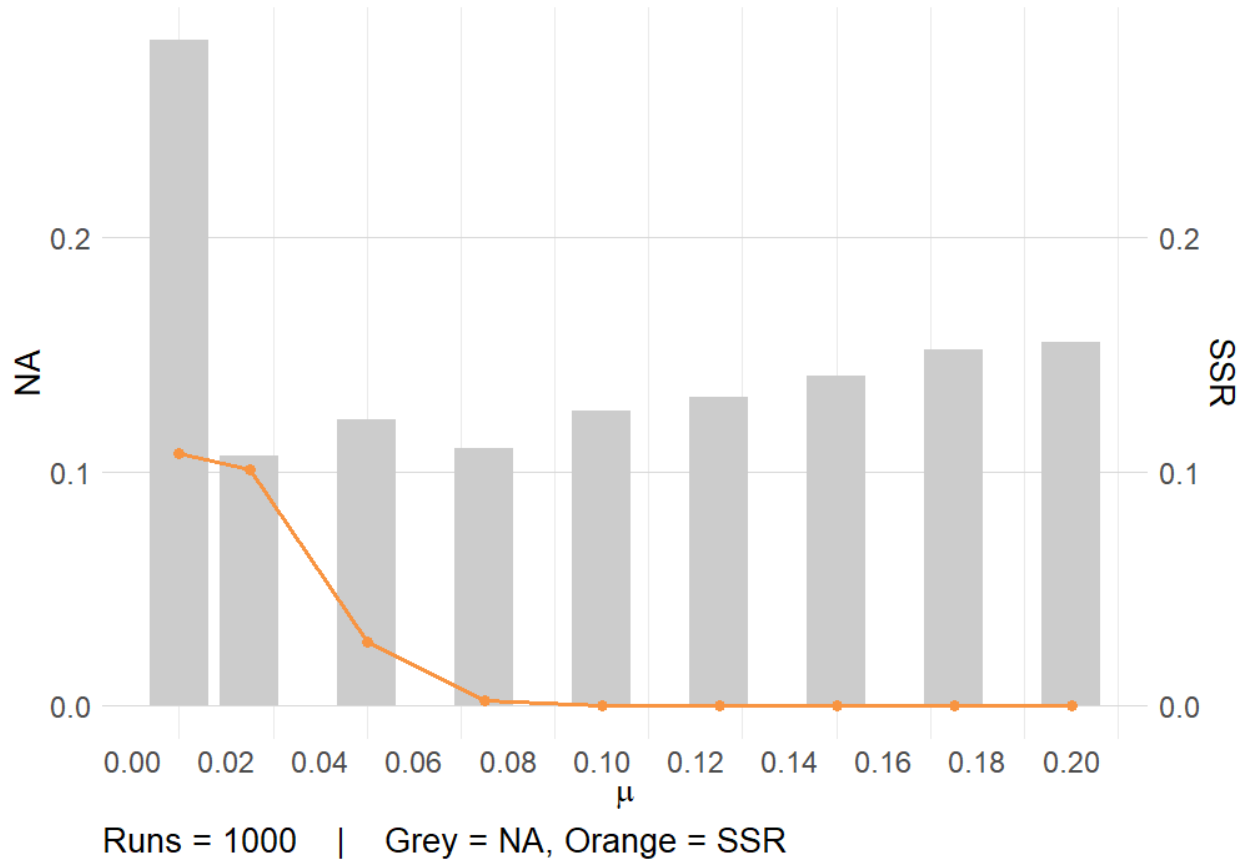


Figure 21: Effect of innovation rate (μ) on SSR and NA under content-biased transmission (snapshot). Higher μ reduces NA and increases detection power, though SSR remains low overall.

To reflect more realistic sampling conditions, we applied the same univariate parameter sweeps using a time averaging scheme. In addition, we swept over time window width (w) to explore the effects of binning on selection detection. As in the previous case, SSR represents the number of selection-flagged variants across runs, and FNR the rate of Type II Error, both excluding of non-analysable variants (NAs). Summary statistics are provided in Tables 7 and 8, and plotted in Figures 25 – 29:

- Innovation rate (μ) revealed a similar negative pattern of SSR to the snapshot version. Departing from 0.2683 at $\mu = 0.01$, SSR dips to ~ 0 at $\mu = 0.10$ onwards (Fig. 25). In addition, %NA spikes to ~ 90 % levels across 1000 runs. μ reveals a greater confounding impact on both selection detection and test analysis when frequencies are aggregated over broader temporal windows.
- Population size (N). As populations become bigger SSR seems to increase notably to ~ 0.95 ; in fact, at $N = 700, 900$ and $1,000$ SSR surpasses the 0.95 power threshold, yielding the highest power levels throughout the content-biased parameter sweeps (Fig. 26). On the other hand,

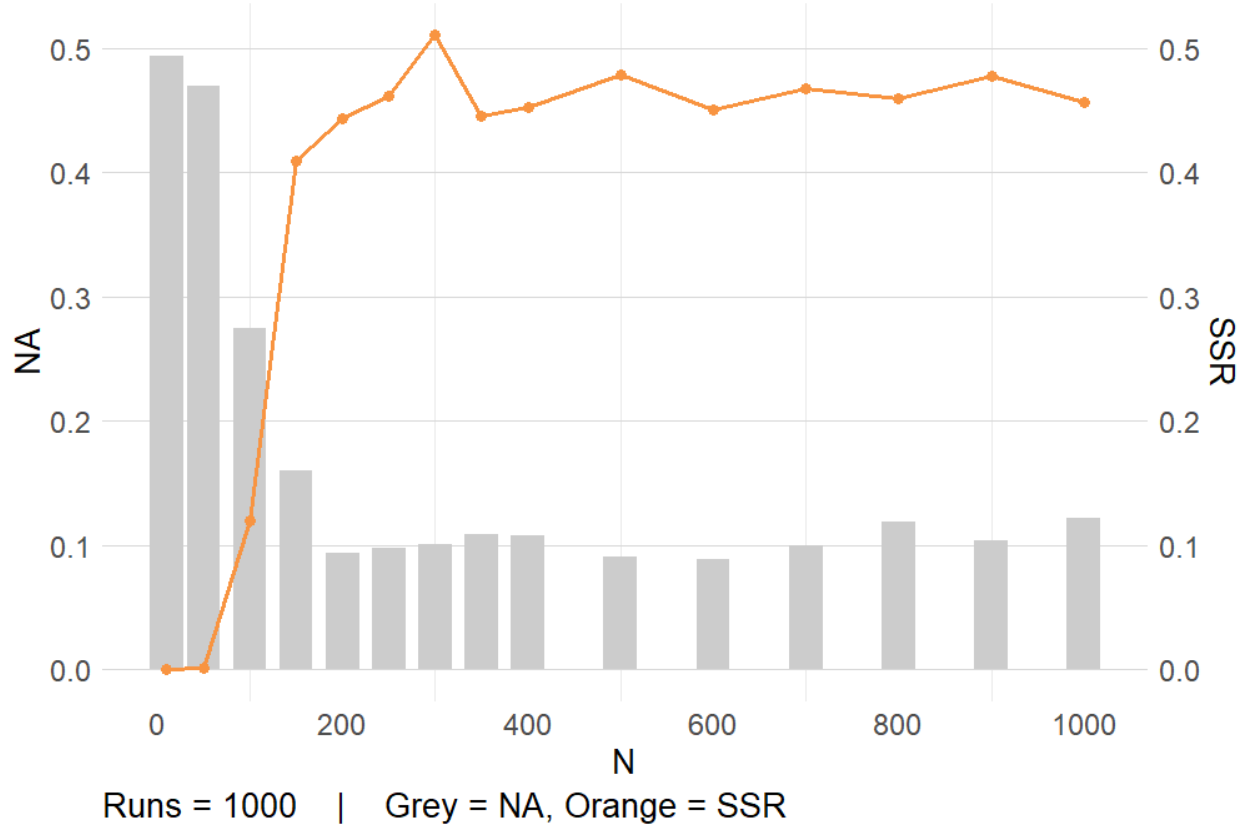


Figure 22: Effect of population size (N) on SSR and NA under content-biased snapshot scheme. SSR increases with N up to ~ 500 , then plateaus. NA decreases as N grows, stabilizing below 15 %.

%NA is consistently aggregated above 50 %. This finding suggests that, although the entire population cannot be analysed, power optimised whenever drift effects are minimised.

- Content bias strength (b) shows a comparable pattern of SSR as the snapshot scheme, although this time it peaks higher, 0.3850 at $b = 0.10$, but it eventually dips to ~ 0 at $b = 0.30$ (Fig. 27). Conversely, %NA reflects an opposite trend to the snapshot scheme, it decreases with stronger biases. Thus, when more variants are analysed, the SST fails to detect them in every instance.
- The number of time steps had a similar critical role under time averaging. Short time series ($t \leq 150$) returns high NA proportions and SSRs, with a noticeable improvement when reaching $t = 50$ (SSR = 0.4153), and a major dip to 0 beyond that (Fig. 28). This suggest that while longer time series keep undetected cases constant, they do not improve the detection of content bias. Thus, intermediate-length series better capture the content bias dynamics.
- Time window width (w). Narrow windows ($w < 20$) resulted in high SSRs (Fig. 29), and proportions of non-detected variants remained stable across sweeps (~ 60 %). We varied t and w simultaneously and discovered that shorter time series and bins better capture selection

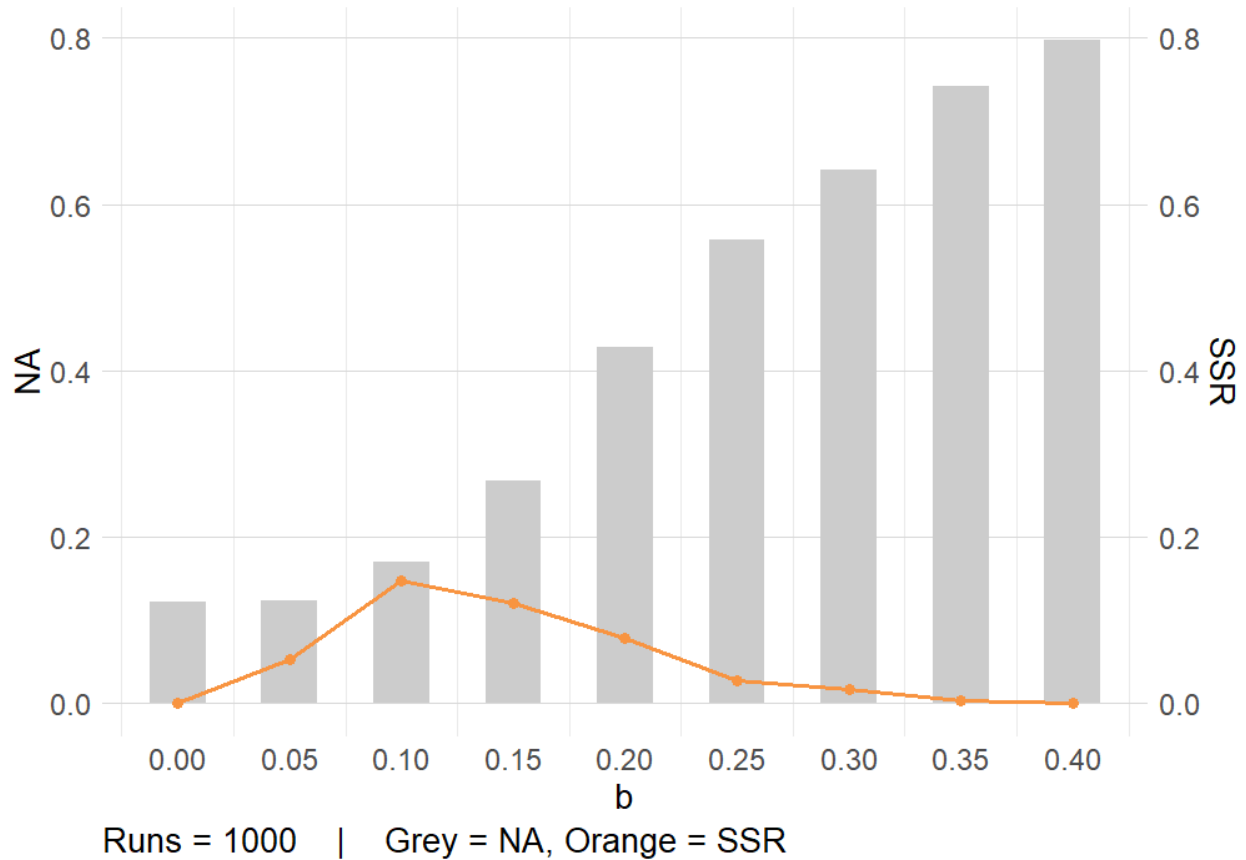


Figure 23: Effect of content bias strength (b) on SSR and NA under snapshot conditions. SSR peaks at $b = 0.1$, but drops as b increases. Stronger bias leads to more missing values (NA up to 80 %).

events. For instance, $w = 10$ and $t = 100$ yielded an intermediate proportion of NAs, but a high SSR (0.7109).

Content-biased transmission revealed important asymmetries and high variability in SST performance under different time schemes. Under snapshot conditions, the test consistently struggled to detect selected variants, with most SSR values falling below the 0.50 threshold across sweeps. Its sensitivity was particularly poor at higher innovation rates and with stronger selection strengths, suggesting interference from variant turnover and early fixation. Conversely, time averaging improved power in several instances, notably in larger populations and with narrower time windows, where SSR occasionally surpassed 0.70, and even the 0.95 threshold. However, this gain came at the cost of high %NA, indicating a trade-off between power and analysability. Overall, the SST performs best under constrained and intermediate conditions, struggling when selection signals are either too strong or too diluted by drift, innovation, or time.

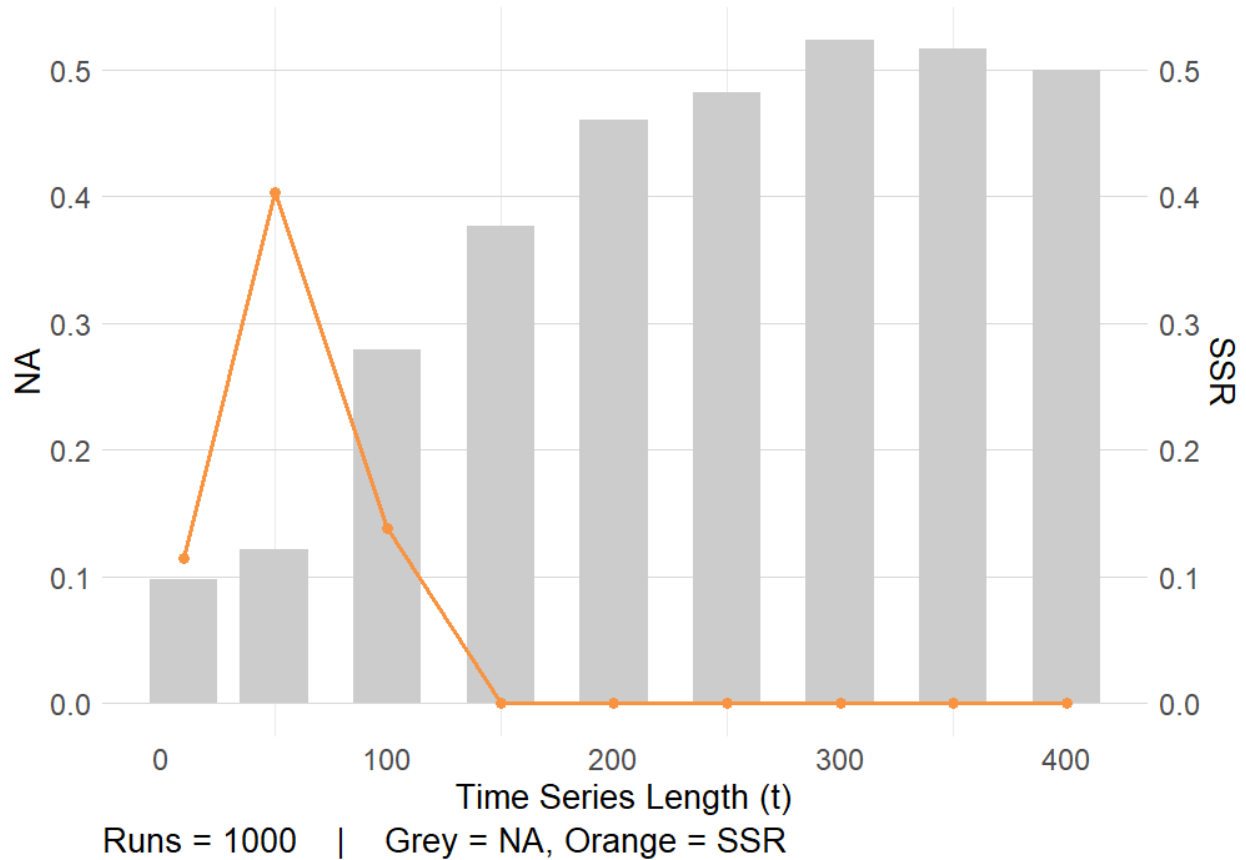


Figure 24: Effect of time series length (t) on SSR and NA under content-biased snapshot. Longer series greatly reduce NA and improve SSR, reaching full detection above $t = 150$.

4.2.1.2. Conformist-biased transmission

Due to the nature of conformist transmission, a complete sweep across the full parameter set was deemed unnecessary. Unlike content-biased selection—where the strength of preference acts directly on the focal variant—conformist bias is determined by the relative frequency of the variant. This mechanism makes the detection of selection at the level of a single focal variant conceptually problematic. As a result, selection pressures in conformist transmission emerge more diffusely, making the SST less appropriate for fine-grained sweeps⁴. Instead, we focused on representative baseline runs, and a univariate sweep of the bias strength parameter (c) and another one of population size (N), both under snapshot and time averaged schemes.

Under baseline parameters ($N = 100$, $\mu = 0.02$, $B = 1000$, $t = 100$, $c = 0.05$, $\alpha = 0.05$, 100 runs) the snapshot model showed modest performance for detecting conformist selection. The

⁴Although not included here, we conducted two parameter interaction sweeps for both versions of the conformist biased transmission model: innovation rate, population size and time series length with conformist bias strength each separately. Scripts available at <https://github.com/Cooperr18/Dissertation-Code>

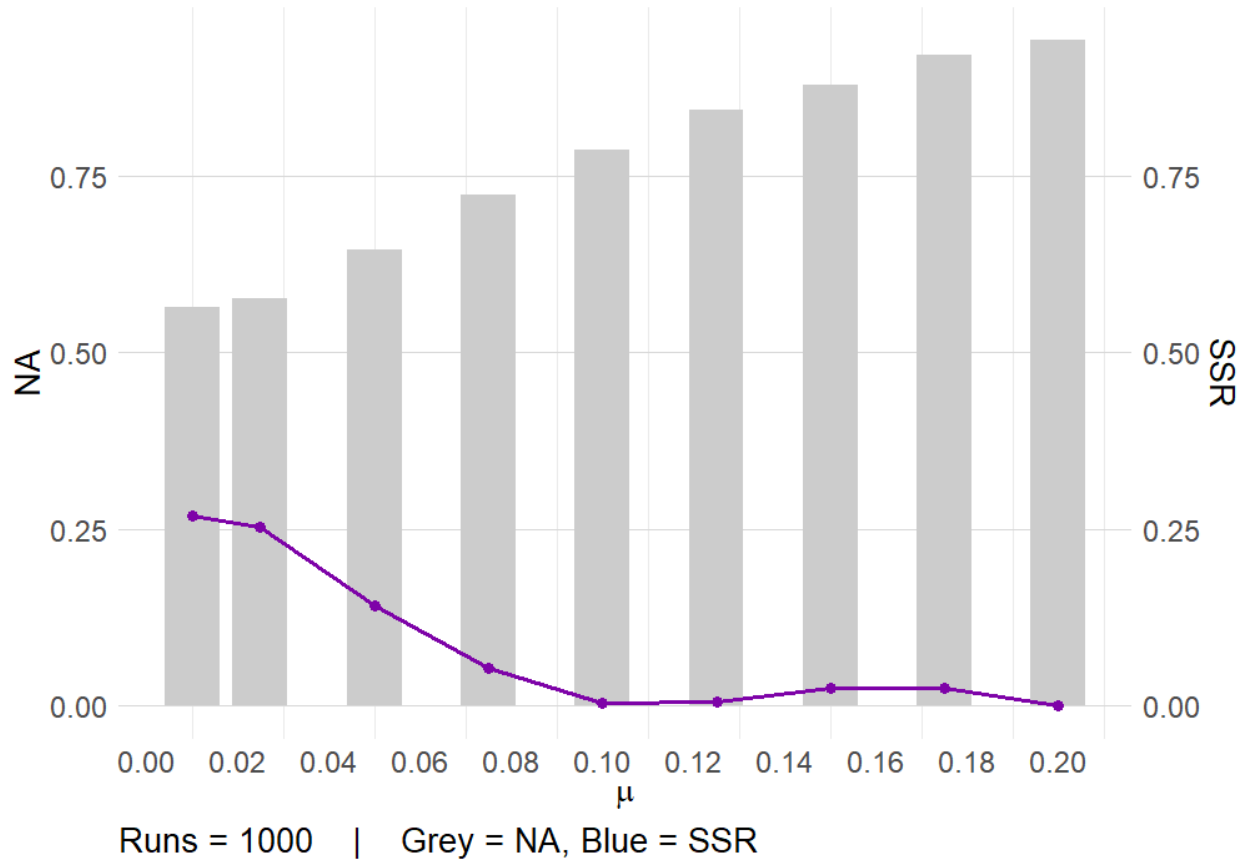


Figure 25: Effect of innovation rate (μ) on SSR and NA under content-biased transmission with time averaging. SSR declines as μ increases, while NA steadily rises.

SSR was just 0.2109, and the proportion of non-analysable variants (%NA) remained at 46.9 %, suggesting the focal variant persisted long enough to be tested, but not to be detected. In the time averaged scenario (100 time steps binned into 10 bins), power deteriorated further, SSR dropping to 0, however, %NA remained at 5 %. These results suggest that the SST fails to flag frequency increases generated through conformist bias, particularly when frequency change is not consistently monotonic.

Baseline results are not very promising, however, when varying population size and bias strength we found that performance is enhanced in some instances. Summary statistics of both snapshot and time averaging versions are provided in Tables 9 and 10, and illustrated in Figures 30 – 33.

- Bias strength (c): Increasing c in the snapshot model showed no systematic improvement in SSR, which hovered close to zero across all values (Fig. 30). Meanwhile, %NA remained high (< 90 %), suggesting the test could not analyse most focal variants, and when it did, it failed to classify them as selected most of the times. This indicates that even mid-range

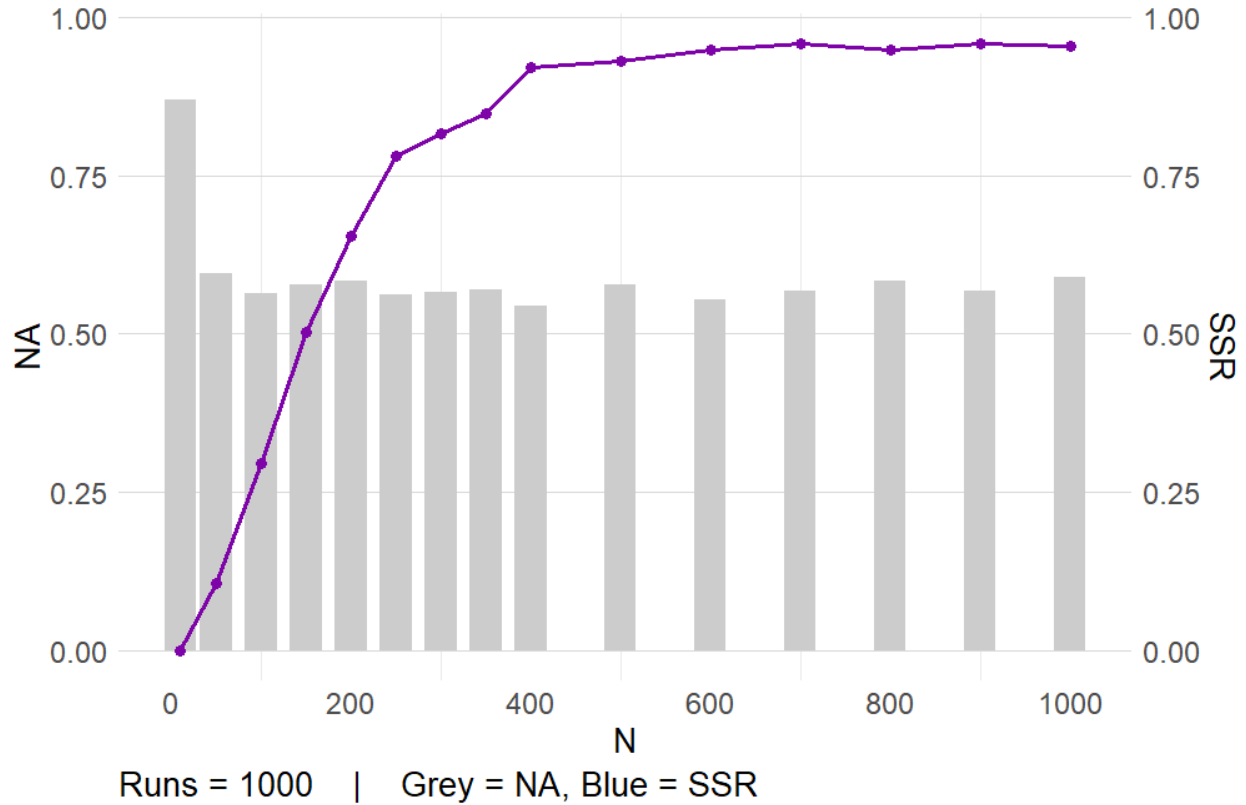


Figure 26: Effect of population size (N) on SSR and NA under content-biased transmission with time averaging. SSR improves rapidly with N , reaching near-perfect detection at $N \approx 400$. NA remains stable around 55–60 %.

conformist dynamics do not generate a consistent directional change that the FIT can detect as selection events.

- Population size (N): Varying N under snapshot conditions produced a clearer pattern (Fig. 31). As population size increased, SSR rose from 0.471 at $N = 20$ to 0.52 at $N = 1,000$, while %NA decreased drastically from 83 % to 3 % in three sweeps. This reflects a clearer frequency signal under larger populations, where drift is reduced and conformity-driven increases become more stable. In this sense, the power under conformity follows a pattern similar to under content bias, implying that overall N is a parameter that enhances the signals of selection even if they processes differ.
- Bias strength (c). Similar to the snapshot model, increasing c under time averaging had minimal impact on SSR. SSR remained flat at 0 throughout the sweep, with %NA values oscillating around 6–8 % (Fig. 32). This reinforces the notion that conformist bias rarely produced the kind of consistent directional frequency shifts needed for the SST to detect selection.

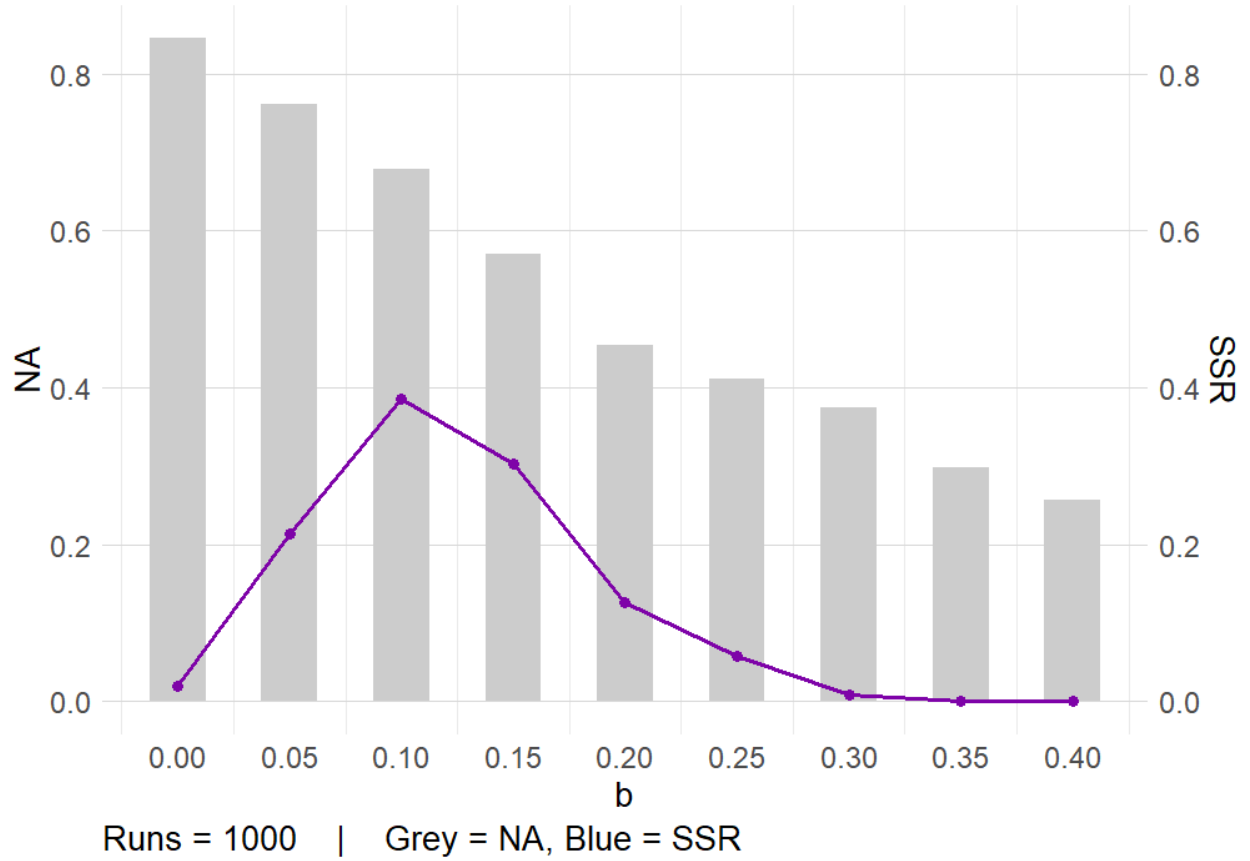


Figure 27: Effect of content bias strength (b) on SSR and NA under content-biased transmission with time averaging. SSR peaks at moderate bias ($b \approx 0.1$), then drops as b increases. Stronger bias also reduces NA.

- Population size (N). Increasing N had little effect on detection rates under time averaging. SSR stayed near zero regardless of N , and %NA remained consistently low (Fig. 33). These findings suggest that while time averaging typically improves analysability by aggregating more observations per time step, it also suppresses short-term variation. This temporal smoothing makes it harder for the SST to detect the subtle and emergent frequency changes driven by conformist bias, further obscuring an already diffuse signal of selection.

The SST consistently failed to detect selection under conformist-biased transmission, regardless of time scheme, bias strength, or population size. In the snapshot model, larger populations improved SSR modestly, but results never approached the conventional power threshold (≥ 0.95). Time averaging further dampened the signal.

These findings reflect a core limitation of the SST: its dependence on monotonic, directional, and drastic frequency increases. Under conformist dynamics the focal variant often reaches high frequency quickly and then stabilises. Thus, the time series lacks the steady upward trend that the

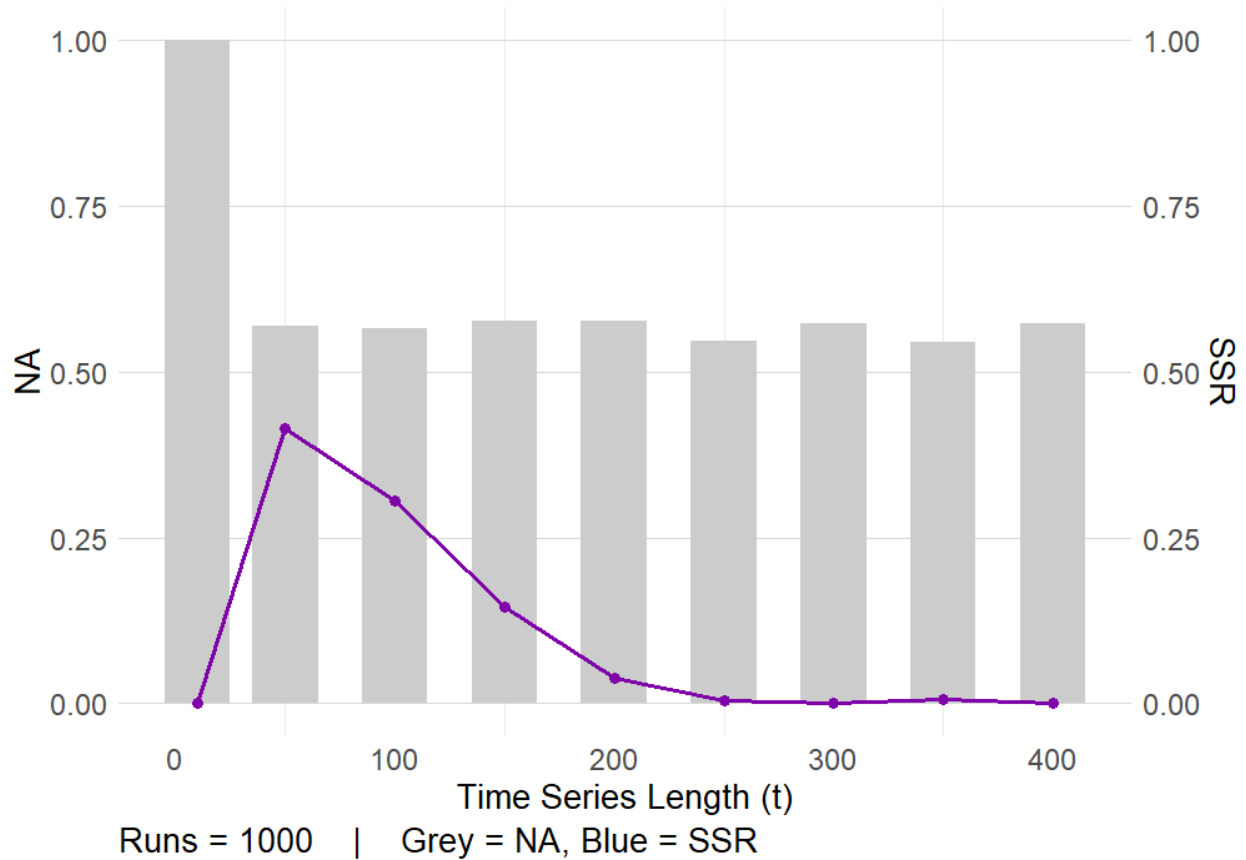


Figure 28: Effect of time series length (t) on SSR and NA under content-biased transmission with time averaging. Longer t lower NA and slightly improve SSR, though detection remains limited.

FIT component of the SST requires to detect selection. As such, even when selection is acting through conformity, the focal variant's early saturation causes the test to miss.

4.2.2. Discussion

Having established the SST's reliability under the null hypothesis, Experiment 2 confronts the ultimate question for its archaeological utility: is the test powerful enough to detect selection when it is actually occurring? This inquiry shifts our focus from the risk of Type I error to the far more common problem of Type II error—the failure to reject the null, i.e. not being able to identify episodes of selection. The results of this experiment are sobering, revealing that the SST's power is deeply contingent and exposes a series of visible trade-offs that strike at the heart of the archaeological inference.

Under the snapshot time scheme the test's power to detect content-biased selection is evidently weak. Over 80 % of genuinely focal variants were misclassified as neutral across sweeps, revealing fundamental limitations when we dispose of fine-grained information about transmission

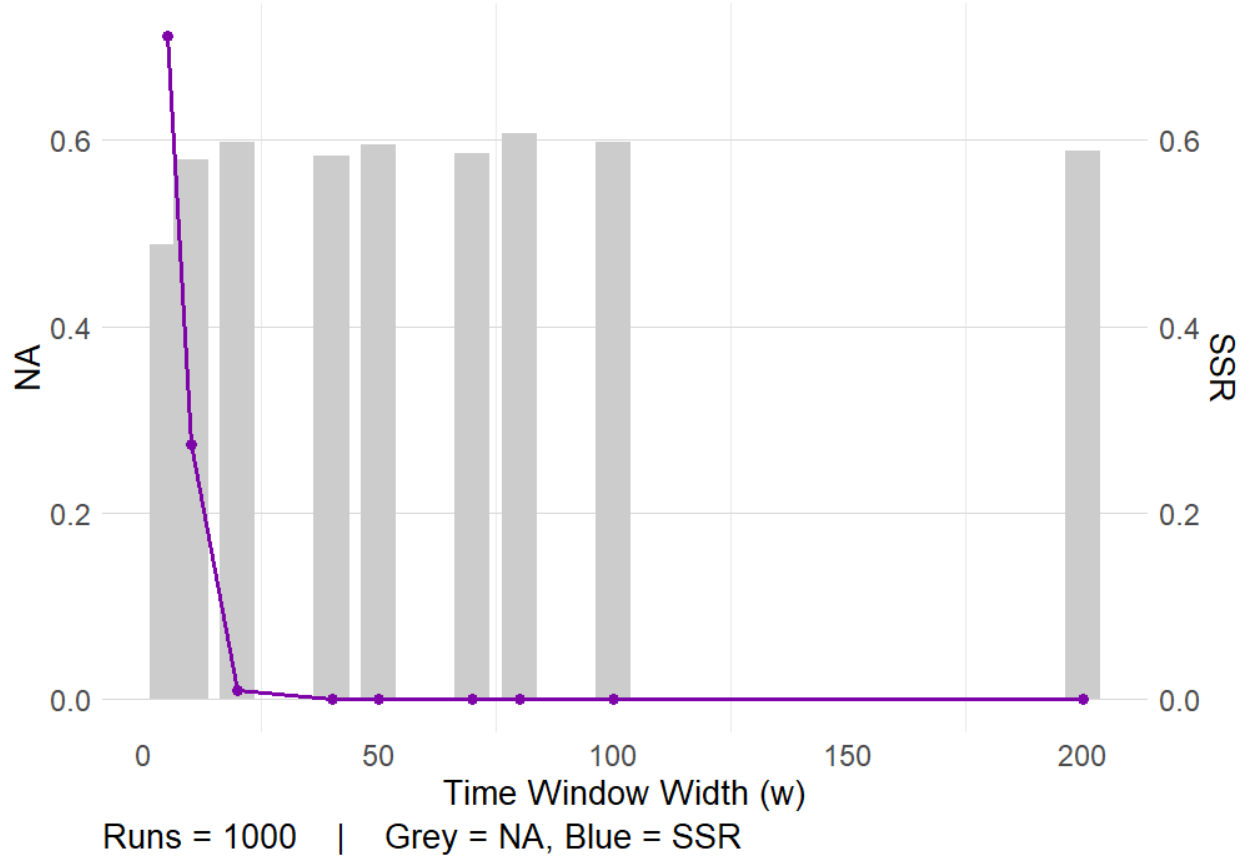


Figure 29: Effect of time window width (w) on SSR and NA under content-biased transmission with time averaging. SSR drops sharply as w increases, while NA remains high beyond small window sizes.

and depositional events. Furthermore, as outlined in the results, the proportion of non-detected variants remains consistently variable.

To better capture what is beyond the signals of selection we must redirect our attention to the model. Selection curves reflect a pattern of high sensitivity for extreme values (Fig. 9, from Feder, Kryazhimskiy and Plotkin (2014)), meaning that at both ends of the curve we obtain the least information about frequency changes and variability. Because in our model the probability of adoption is given by the frequency of the variant and b fundamentally (Eq. 5), fluctuations must vary as a function of these parameters. In the content-biased model we observe a pattern analogous to the coefficient of selection curve (Fig. 9, from Feder, Kryazhimskiy and Plotkin (2014)): when content bias is too strong the focal variant reaches fixation before $t = 10$, so that it fluctuates around 0.90 frequency for the rest of the time series, yielding insignificant information about variability; on the other hand, when b strength is weak drift overwhelms selection and the focal variant goes extinct before succeeding for three time points. Thus, it is within mid-range values that power is optimised.

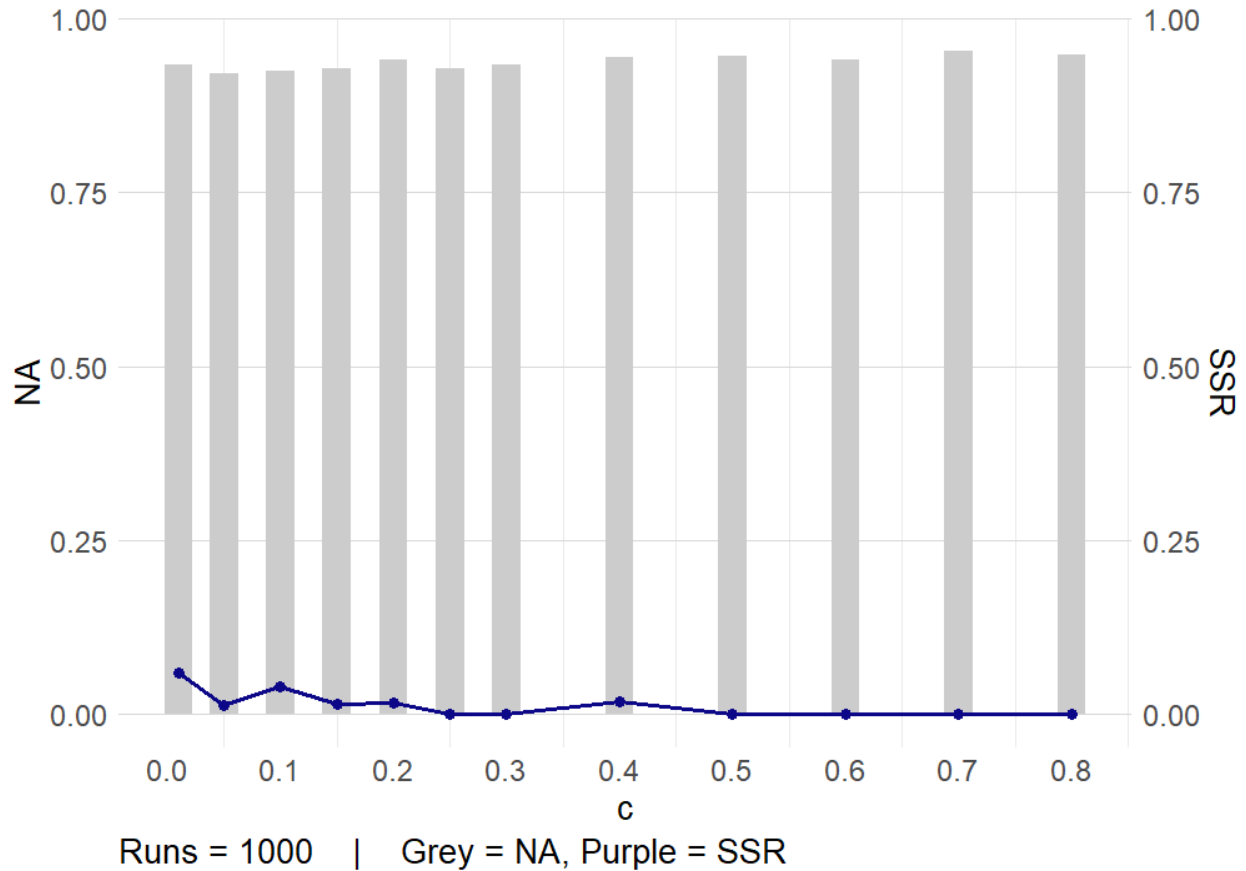


Figure 30: Effect of conformist bias strength (c) on SSR and NA under snapshot conditions. SSR remains low across all values of c , with NA consistently high ($\approx 90\%$).

Controlling for b did not result in increased SSR nor smaller %NA, but in fact population size that did it. With increasing N , SST's power increased noticeably (Fig. 22), mainly because the effects of drift were reduced. The step-by-step randomness provoked by cultural drift faded away with higher population sizes, even when keeping b constant across sweeps; N “unmasked” this complex relationship.

This picture creates a classic case of equifinality, where distinct processes such as cultural drift and weak/strong selection biases produce statistically indistinguishable patterns. This is a critical theoretical concern raised throughout the literature (Premo, 2010; Rorabaugh, 2014; Barrett, 2019; Madsen, 2020), who caution against the naïve interpretation of statistical outputs without a deep understanding of the processes that generate them. Our findings provide a stark, quantitative illustration of this principle. Even under ideal conditions, discriminating between neutral and selective processes proves extraordinarily difficult, suggesting that the challenge could be at the nature of cultural evolutionary dynamics.

When we transition to the time averaging scheme the test demonstrated improved power

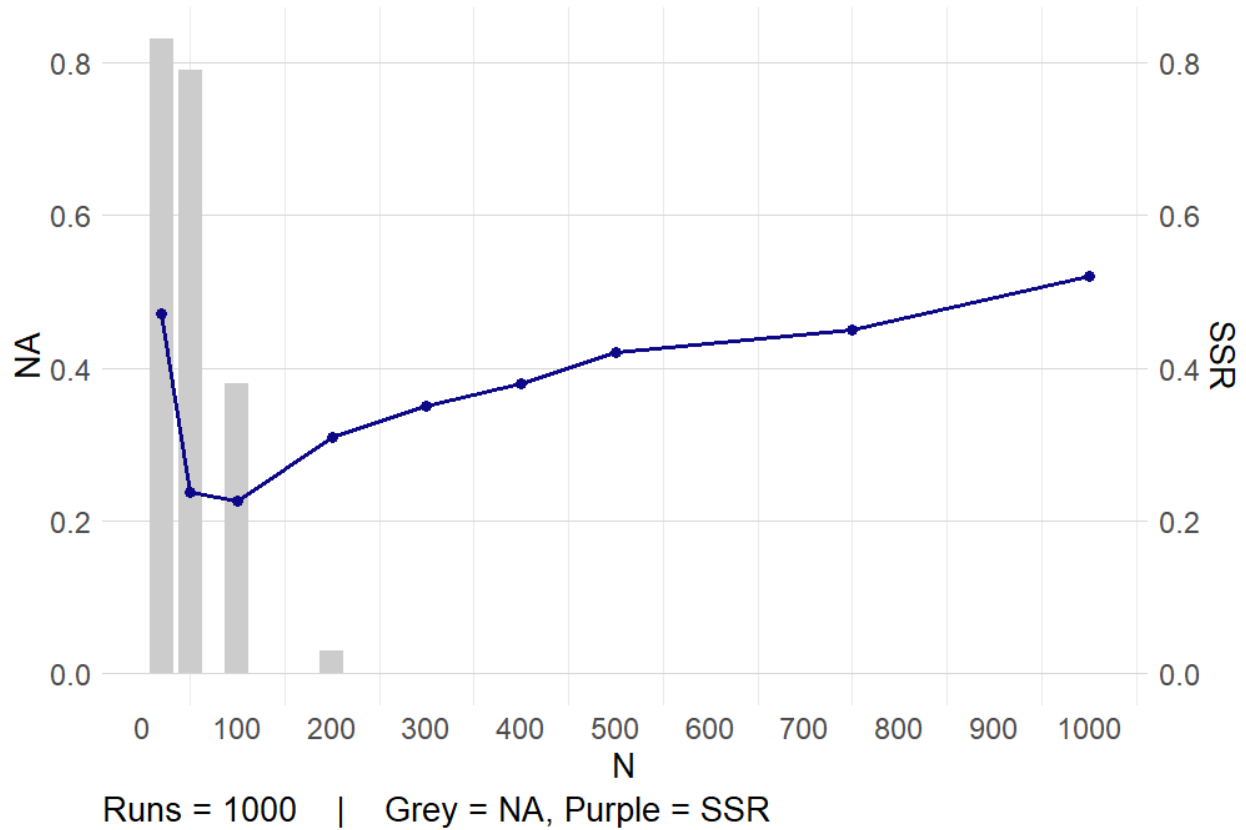


Figure 31: Effect of population size (N) on SSR and NA under conformist bias (snapshot). SSR increases with N , reaching ~ 0.5 at $N = 1000$. NA decreases initially but stays above 20 %.

overall. The mechanism for this increased sensitivity is intuitive: by averaging out high-frequency stochastic fluctuations across temporal bins, the persistent signals of selection become a more dominant and statistically detectable trend. However, this gain is achieved through a major trade-off: non analysable variants skyrocket to ~ 50 % overall. This means that in exchange for detecting selection in some variants we render more than half of them completely invisible to analysis.

Our results shed light on the mechanism responsible for this data loss: stronger selection pressures cause variants to reach fixation or extinction within a single time averaged bin. The variant's trajectory—injection, rise and potential fixation—is compressed into a single temporal bin. Because the “three-time-point rule” requires a variant to be present and changing across multiple analytical units, the early success of a variant ensures its analytical erasure. This paradox reflects the deeper problem noted by Feder, Kryazhimskiy and Plotkin (2014), whose solution is again found controlling for population size (Figs. 23 and 26). As the simulations demonstrate, when time series are shortened or when variants experience rapid frequency changes, the test loses the temporal information necessary for accurate inference. However, when given a fairly large population size, the effects of drift are dissipated, and the focal variant can gradually rise.

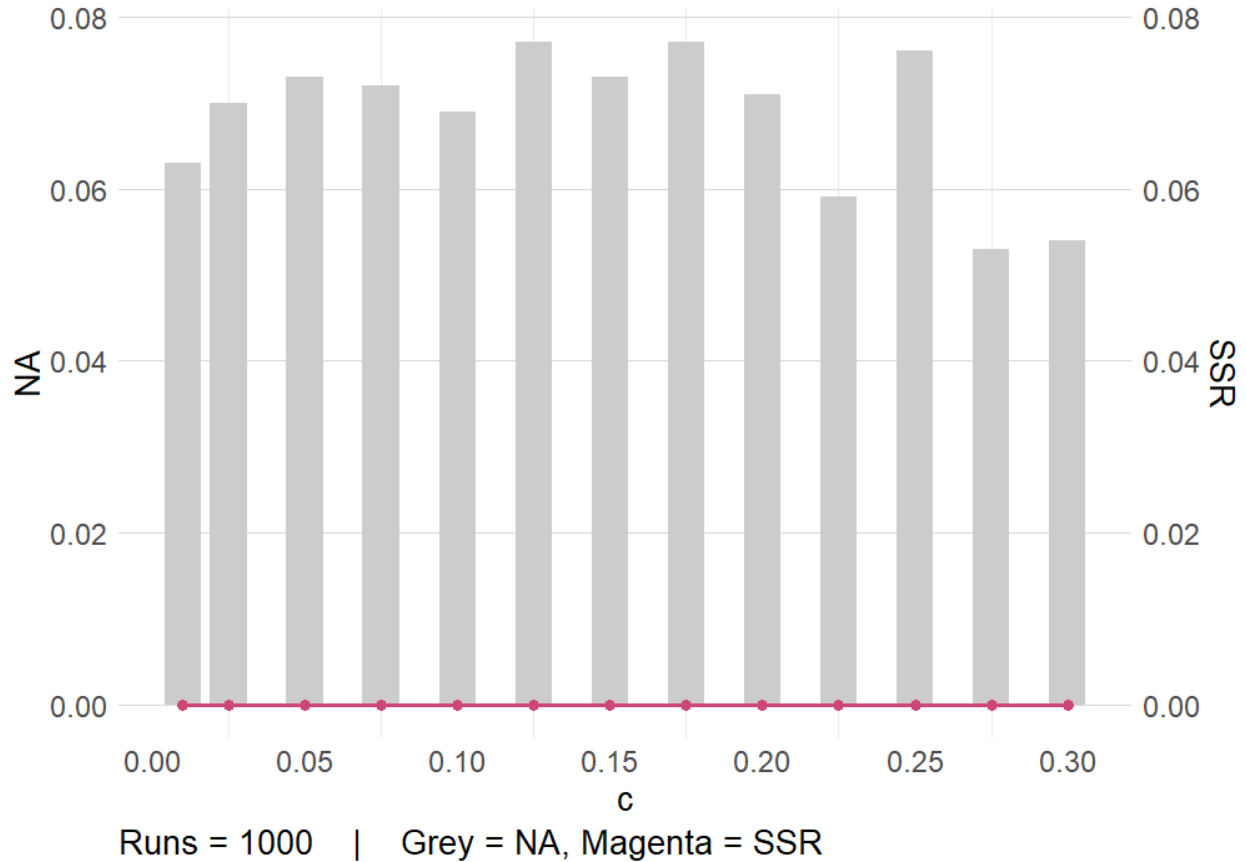


Figure 32: Conformist bias under time averaging: effect of bias strength (c). SSR remains at zero across all values, while NA remains uniformly high.

This finding places archaeologists in a methodological bind that mirrors broader challenges in the discipline. Time averaging, while making analysis possible by aggregating sparse archaeological data, simultaneously obscures the very processes of change we wish to study (Premo, 2014). This dilemma is not unique to the SST, as other authors have highlighted the property of time averaging to mask complex interactions of cultural transmission (Madsen, 2012, 2020; Premo 2014; Porčić, 2015; Crema, Kandler and Shennan, 2016). It simply reflects a fundamental tension in archaeological inference between temporal resolution and analytical tractability, to which we must add more layers of complexity such as spatial scale and the boundaries of the units of transmission (Lipo et al., 1997; Eerkens and Lipo, 2005; O’Brien and Shennan, 2010; Premo and Scholnick, 2011).

The SST’s limitations become even more pronounced when we consider conformist-biased transmission. Here, the test fails almost entirely regardless of the time scheme and parameter sweep employed. Under snapshot conditions even large populations yield modest SSR values that never come close to the conventional power threshold (Tab. 9). Time averaging, which proved beneficial for content bias, further suppresses the already weak conformist signal (Tab. 10). The reason lies in

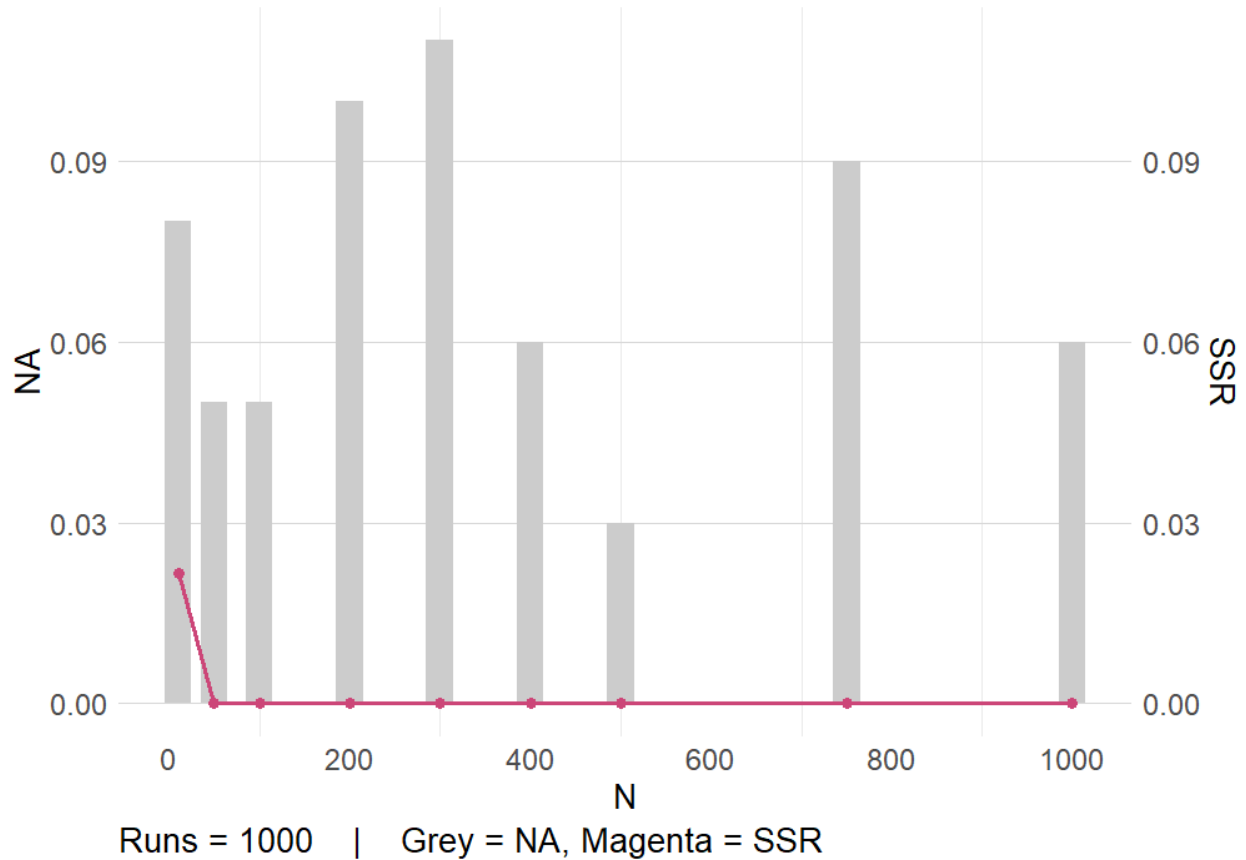


Figure 33: Conformist bias under time averaging: effect of population size (N). SSR remains at zero, and NA fluctuates slightly but stays above 6 %.

the nature of conformist transmission: it often produces frequency-dependent dynamics that may not generate the consistent directional changes the SST is designed to detect. When popular variants become more popular, the resulting frequency trajectories become too subtle or too variable to be discriminated from neutral drift.

This caveat represents more than a statistical limitation; it points to a fundamental mismatch between the test's design assumptions and the variability produced by cultural transmission mechanisms. Cultural evolution encompasses a much broader range of processes, including mixed learning, guided variation, anti-conformist and pro-novelty biases, prestige effects, spatial transmission patterns, artefact constraints, copying errors, just to name a few (Mesoudi, 2011). These could easily not conform to the test's underlying assumptions, which is an issue worth pondering in future studies. This limitation echoes broader critiques of applying population genetic models to cultural data without adequate modification for the unique properties of cultural transmission (Kandler and Crema, 2019; Madsen, 2020).

It is also yet to explore if these limitations are dissipated when varying the number and

replicates of focal variants whose transmission is biased ($b/c > 0$). After varying the number of replicates at $t = 1$ throughout the design of the experiment, I decided to stick to just adding a single count to maximise analytical simplicity⁵. We have not attempted to introduce more than one focal variant for the same reason, but we encourage future researchers to carry out such task. Nonetheless, we argue that adding more replicates would accrue the caveat outlined above: with higher counts less time until fixation is required, always limited to smaller populations. Additionally, adding more focal variants ($b/c > 0$) could perhaps limit FIT's tractability and blur the outcomes.

We should further acknowledge the scope and limitations of this study regarding the SST and FIT's original formulation by Feder, Kryazhimskiy and Plotkin (2014). Firstly, we did not explore the test's ability to estimate the coefficient of selection (s), a primary focus of the original work. While Feder and colleagues provided extensive analysis on genetic data, our study models election via a content bias model, focusing on binary detection outcomes. This leaves open critical questions about quantifying selection strength from archaeological data—a mandatory step for moving beyond p -values to capture the true dimension of selective forces (Wasserstein, Schirm and Lazar, 2019). Secondly, our analysis did not cover the issue of sampling intervals. Feder, Kryazhimskiy and Plotkin dedicated substantial effort to calculate the optimal sampling intervals where power is maximised, and we suspect it is a matter worth studying with archaeological data. Despite this, we opted not to cover sampling intervals due to time limits and the need to establish simplified, controlled conditions that approximate archaeological realities without introducing additional analytical complexity⁶. Nonetheless, a comprehensive evaluation of discrete sampling strategies remains essential for providing practical guidance to archaeological research design in the future.

To sum up, this experiment demonstrates that the SST is not a simple “test of selection” but a complex instrument whose effectiveness is highly contingent on the strength of biases, the nature of the transmission processes, and the structure of the temporal data and the populations. The snapshot scheme provided a useful theoretical baseline, revealing fundamental challenges posed by equifinality. Time averaging introduced its own trade-offs between detection power and population testability. These findings serve as cautionary evidence for the application of quantitative methods in archaeology. We here emphasise that statistical tools must be understood not as black boxes, but as instruments whose limitations are as important as their possibilities.

⁵See “Report of daily progress” at <https://github.com/Cooperr18/Dissertation-Code> to browse the results of varying initial count.

⁶In the “Report of daily progress” there is preliminary application of the OSI (Optimal Sampling Interval) scheme, with a special focus on minimising %NA rather than maximising SSR. Code and some plots available at <https://github.com/Cooperr18/Dissertation-Code>.

Chapter 5. Conclusion

This dissertation sought to answer a foundational question for evolutionary archaeology: can we reliably discriminate the signals of selection from the noise of drift within the fragmentary and scarce archaeological data? The answer, as we have demonstrated, is not a simple affirmative or negative. Instead, we affirm that it is a rather complex and conditional *yes*, bounded by methodological trade-offs and theoretical limitations. This investigation deconstructed the Signal Selection Test and its core statistics, the Frequency Increase Test, moving beyond a simple application to evaluate its performance under controlled, simulated experiments. The findings indicate that while the SST can be a valuable component of an archaeologist's toolkit, its use requires a profound understanding of its biases and a critical awareness of the paradoxes it creates.

The core finding of this investigation is that the SST's utility is highly conditional, defined by a tension between reliability, statistical power and evidence preservation. While the test is robust against Type I error, its power to detect genuine selection is significantly limited. The test's ability to identify content-biased selection is compromised by a trade-off: the time averaging required to amplify the signal simultaneously erases the majority of informative variants from the analysis due to rapid fixation. In addition, the test proved entirely blind to conformist transmission, showcasing a critical design flaw that makes it unsuited for detecting this mechanism of cultural transmission as we modelled it. We encourage future researchers to conduct similar analyses using different approaches to biased transmission.

In sum, we have demonstrated that the Signal Selection Test is a complex tool whose output is deeply contingent on the interplay between the mode of transmission, the strength of selection, and the structure of the populations and time. It is yet to explore the nature of this interplay when we consider other outputs of the SST, such as initial frequency, estimated population size and estimated coefficient of selection (s), and sampling intervals (Feder, Kryazhimskiy and Plotkin, 2014; Newberry et al., 2017). Additionally, different methodological approaches could potentially lead to improved results, such as testing through samples of population (Kandler and Shennan, 2013), applying mixed, complex transmission models (Crema, Kandler and Shennan, 2016), the boundaries of the units of transmission (Crema, Bortolini and Lake, 2024; Premo and Rezek, 2024), and what limits the motors of social learning, guided variation and innovation (Boyd and Richerson, 1985; Henrich, 2001; Eerkens and Lipo, 2005; Legare and Nielsen, 2015).

In sum, the significance of this project is threefold. First, it provides a stark and quantitative demonstration of equifinality, showing how easily weak and strong selection can be mistaken for neutral evolution in archaeological data. Second, it highlights the profound challenge that time averaging poses, not just as a taphonomic issue, but as a source of systematic analytical bias (O'Brien,

2003; Madsen, 2012; Premo, 2014; Porčić, 2015). Third, it underscores the limitations of applying population genetic models directly to the archaeological record without accounting for its unique properties (Bentley, Hahn and Shennan, 2004; Eerkens and Lipo, 2005; Shennan, 2011; Kandler and Shennan, 2015; Kandler and Crema, 2019). The ultimate implication is a call for methodological pluralism, where we suggest the SST not to be used in isolation, but as a component within a suite of inferential tools, an inferential “toolbox”. Ultimately, it is this critical understanding of each tool’s limitations that allows us to confidently distinguish between genuine evolutionary signals and stochastic dynamics.

Appendix

FIT application and likelihood-based s and population size estimations are available at <https://github.com/bacovcin/tsinfer-R> in R language; the application to the LBK Merzbach ceramic assemblage at <https://github.com/benmarwick/signatselect>. Both repos are publicly accessible.

We do not analyse or generate any datasets, because this work proceeds within a theoretical and simulated approach. The code to replicate the experiments' results, figures is available at <https://github.com/Cooperr18/Dissertation-Code>. Additionally, the repo includes figures and tables not used here, along with a daily report to provide full transparency of the decisions made and discarded along the realisation of the project.

The application of the FIT to the cultural transmission models is guided by Marwick et al.'s repo, although the core code of the models is borrowed from Crema et al. (2016), R code available at <https://github.com/ercrema/CulturalTransmissionModel>. Thus, much credit is given to my supervisor Dr Enrico Crema, who really gave me the little push I needed in the beginning.

Tables

BASELINE							
N	μ	B	t	NDR	FPR	%NA	Runs
100	0.02	1000	1000	0.998319	0.001681	17.55064	100
<i>Varying μ</i>							
100	0.01	1000	1000	0.997797	0.002203	17.02227	1000
100	0.025	1000	1000	0.998874	0.001326	17.664	1000
100	0.05	1000	1000	0.999026	0.000971	18.6664	1000
100	0.075	1000	1000	0.999171	0.000829	19.70491	1000
100	0.1	1000	1000	0.999208	0.000792	20.76235	1000
100	0.125	1000	1000	0.998727	0.000782	21.67456	1000
100	0.15	1000	1000	0.998271	0.000729	23.03507	1000
100	0.175	1000	1000	0.999504	0.000716	24.27196	1000
100	0.2	1000	1000	0.998511	0.000690	25.52257	1000
<i>Varying N</i>							
10	0.01	1000	1000	0.998594	0.001406	35.24117	1000
50	0.01	1000	1000	0.998967	0.001933	17.86028	1000
100	0.01	1000	1000	0.997798	0.002202	17.00167	1000
150	0.01	1000	1000	0.997713	0.002287	16.93192	1000
200	0.01	1000	1000	0.997639	0.002341	17.03137	1000
250	0.01	1000	1000	0.997677	0.002633	16.96283	1000
300	0.01	1000	1000	0.997653	0.002267	16.98300	1000
350	0.01	1000	1000	0.997694	0.002306	17.00312	1000
400	0.01	1000	1000	0.997697	0.002303	17.00813	1000
<i>Varying t</i>							
100	0.01	100	100	0.98333	0.01667	15.61518	1000
100	0.01	200	200	0.99134	0.00866	16.33554	1000
100	0.01	500	500	0.99627	0.000373	16.79992	1000
100	0.01	750	750	0.997517	0.002855	16.98859	1000
100	0.01	1000	1000	0.997677	0.002323	17.07776	1000
100	0.01	1500	1505	0.99841	0.00159	17.12249	1000
100	0.01	2000	2000	0.998596	0.001404	17.11453	1000

Table 3: Unbiased transmission under snapshot sampling. Neutral Detection Rate (NDR), False Positive Rate (FPR), and proportion of non-assessable variants (%NA) for single parameter sweeps of innovation rate (μ), population size (N), and observation window (t).

BASELINE								
N	μ	B	t	w	NDR	FPR	%NA	Runs
100	0.02	1000	1000	20	0.996056	0.003944	0	100
<i>Varying μ</i>								
100	0.01	1000	1000	20	0.993085	0.006915	0	1000
100	0.025	1000	1000	20	0.99646	0.00354	0	1000
100	0.05	1000	1000	20	0.997008	0.002992	0.000764	1000
100	0.075	1000	1000	20	0.997444	0.002556	0.000858	1000
100	0.1	1000	1000	20	0.997783	0.002217	0	1000
100	0.125	1000	1000	20	0.99819	0.00181	0	1000
100	0.15	1000	1000	20	0.998674	0.001326	0	1000
100	0.175	1000	1000	20	0.998864	0.001136	0	1000
100	0.2	1000	1000	20	0.999446	0.000554	0	1000
<i>Varying N</i>								
10	0.01	1000	1000	20	0.99619	0.00381	28.54363	1000
50	0.01	1000	1000	20	0.993912	0.006088	0	1000
100	0.01	1000	1000	20	0.993152	0.006848	0	1000
150	0.01	1000	1000	20	0.993319	0.006681	0	1000
200	0.01	1000	1000	20	0.993569	0.006431	0	1000
250	0.01	1000	1000	20	0.993452	0.006548	0	1000
300	0.01	1000	1000	20	0.993268	0.006732	0	1000
350	0.01	1000	1000	20	0.993518	0.006482	0.000532	1000
400	0.01	1000	1000	20	0.993427	0.006573	0	1000
<i>Varying t</i>								
100	0.01	100	100	20	0.953119	0.046881	0	1000
100	0.01	200	200	20	0.973008	0.026992	0	1000
100	0.01	500	500	20	0.98926	0.01074	0	1000
100	0.01	750	750	20	0.991006	0.008994	0	1000
100	0.01	1000	1000	20	0.992953	0.007047	0	1000
100	0.01	1500	1500	20	0.994951	0.005049	0	1000
100	0.01	2000	2000	20	0.995663	0.004337	0	1000
100	0.01	2500	2505	20	0.996532	0.003468	0	1000
100	0.01	3000	3000	20	0.996803	0.003197	0.000599	1000
<i>Varying w</i>								
100	0.01	1000	1000	5	0.99318	0.00682	0.0189	1000
100	0.01	1000	1000	10	0.993545	0.006455	0.001851	1000
100	0.01	1000	1000	20	0.993156	0.006844	0	1000
100	0.01	1000	1000	40	0.993014	0.006986	0	1000
100	0.01	1000	1000	50	0.993155	0.006845	0	1000
100	0.01	1000	1000	70	0.993951	0.006049	0	1000
100	0.01	1000	1000	80	0.992307	0.007693	0	1000
100	0.01	1000	1000	100	0.994765	0.005235	0	1000
100	0.01	1000	1000	200	0.996053	0.003947	0	1000

Table 4: Neutral drift under time-averaged sampling ($w = 20$). NDR, FPR, and %NA for single-parameter sweeps of innovation rate (μ), population size (N), observation window (t), and sampling window (w).

BASELINE								
N	μ	b	B	t	SSR	FNR	%NA	Runs
100	0.02	0.15	1000	100	0.181818	0.818182	0.12	100
<i>Varying μ</i>								
100	0.01	0.15	1000	100	0.107542	0.892458	0.284	1000
100	0.025	0.15	1000	100	0.100784	0.899216	0.107	1000
100	0.05	0.15	1000	100	0.027335	0.972665	0.122	1000
100	0.075	0.15	1000	100	0.002247	0.997753	0.11	1000
100	0.1	0.15	1000	100	0	1	0.126	1000
100	0.125	0.15	1000	100	0	1	0.132	1000
100	0.15	0.15	1000	100	0	1	0.141	1000
100	0.175	0.15	1000	100	0	1	0.152	1000
100	0.2	0.15	1000	100	0	1	0.155	1000
<i>Varying N</i>								
10	0.01	0.15	1000	100	0	1	0.494	1000
50	0.01	0.15	1000	100	0.001887	0.998113	0.47	1000
100	0.01	0.15	1000	100	0.12	0.88	0.275	1000
150	0.01	0.15	1000	100	0.409524	0.590476	0.16	1000
200	0.01	0.15	1000	100	0.443709	0.556291	0.094	1000
250	0.01	0.15	1000	100	0.461197	0.538803	0.098	1000
300	0.01	0.15	1000	100	0.510567	0.489433	0.101	1000
350	0.01	0.15	1000	100	0.445567	0.554433	0.109	1000
400	0.01	0.15	1000	100	0.452915	0.547085	0.108	1000
500	0.01	0.15	1000	100	0.478548	0.521452	0.091	1000
600	0.01	0.15	1000	100	0.450055	0.549945	0.089	1000
700	0.01	0.15	1000	100	0.467778	0.532222	0.1	1000
800	0.01	0.15	1000	100	0.459705	0.540295	0.119	1000
900	0.01	0.15	1000	100	0.477679	0.522321	0.104	1000
1000	0.01	0.15	1000	100	0.45672	0.54328	0.122	1000
<i>Varying t</i>								
100	0.01	0.15	1000	10	0.114191	0.885809	0.098	1000
100	0.01	0.15	1000	50	0.40273	0.59727	0.121	1000
100	0.01	0.15	1000	100	0.138696	0.861304	0.279	1000
100	0.01	0.15	1000	150	0	1	0.377	1000
100	0.01	0.15	1000	200	0	1	0.46	1000
100	0.01	0.15	1000	250	0	1	0.482	1000
100	0.01	0.15	1000	300	0	1	0.523	1000
100	0.01	0.15	1000	350	0	1	0.516	1000
100	0.01	0.15	1000	400	0	1	0.5	1000

Table 5: Content bias detection under snapshot sampling. Signal-of-Selection Rate (SSR), False Negative Rate (FNR), and proportion of variants excluded (%NA) from FIT across parameter sweeps: innovation rate (μ), population size (N), and total timesteps (t).

Content Bias Strength Sweep								
N	μ	b	B	t	SSR	FNR	%NA	Runs
100	0.01	0	1000	100	0.001139	0.998861	0.122	1000
100	0.01	0.05	1000	100	0.053653	0.946347	0.124	1000
100	0.01	0.1	1000	100	0.148193	0.851807	0.17	1000
100	0.01	0.15	1000	100	0.120055	0.879945	0.267	1000
100	0.01	0.2	1000	100	0.078809	0.921191	0.429	1000
100	0.01	0.25	1000	100	0.027149	0.972851	0.558	1000
100	0.01	0.3	1000	100	0.01676	0.98324	0.642	1000
100	0.01	0.35	1000	100	0.003876	0.996124	0.742	1000
100	0.01	0.4	1000	100	0	1	0.797	1000

Table 6: Content bias detection under snapshot scheme. Signal-of-Selection Rate (SSR), False Negative Rate (FNR), and proportion of variants excluded (%NA) from FIT across content bias strength (b) sweeps.

BASELINE									
N	μ	b	B	t	w	SSR	FNR	%NA	Runs
100	0.01	0.15	1000	100	10	0.457	0.543	0.54	100
<i>Varying μ</i>									
100	0.01	0.15	1000	100	10	0.268349	0.731651	0.564	1000
100	0.025	0.15	1000	100	10	0.252955	0.747045	0.577	1000
100	0.05	0.15	1000	100	10	0.141243	0.858757	0.646	1000
100	0.075	0.15	1000	100	10	0.054152	0.945848	0.723	1000
100	0.1	0.15	1000	100	10	0.004695	0.995305	0.787	1000
100	0.125	0.15	1000	100	10	0.006410	0.993590	0.844	1000
100	0.15	0.15	1000	100	10	0.025000	0.975000	0.880	1000
100	0.175	0.15	1000	100	10	0.025316	0.974684	0.921	1000
100	0.2	0.15	1000	100	10	0.000000	1.000000	0.942	1000
<i>Varying N</i>									
10	0.01	0.15	1000	100	10	0.000000	1.000000	0.869	1000
50	0.01	0.15	1000	100	10	0.106173	0.893827	0.595	1000
100	0.01	0.15	1000	100	10	0.295195	0.704805	0.563	1000
150	0.01	0.15	1000	100	10	0.502370	0.497630	0.578	1000
200	0.01	0.15	1000	100	10	0.654676	0.345324	0.583	1000
250	0.01	0.15	1000	100	10	0.780822	0.219178	0.562	1000
300	0.01	0.15	1000	100	10	0.816092	0.183908	0.565	1000
350	0.01	0.15	1000	100	10	0.848837	0.151163	0.570	1000
400	0.01	0.15	1000	100	10	0.921053	0.078947	0.544	1000
500	0.01	0.15	1000	100	10	0.931280	0.068720	0.578	1000
600	0.01	0.15	1000	100	10	0.948546	0.051454	0.553	1000
700	0.01	0.15	1000	100	10	0.958430	0.041570	0.567	1000
800	0.01	0.15	1000	100	10	0.949519	0.050481	0.584	1000
900	0.01	0.15	1000	100	10	0.958333	0.041667	0.568	1000
1000	0.01	0.15	1000	100	10	0.953771	0.046229	0.589	1000
<i>Varying t</i>									
100	0.01	0.15	1000	10	10	0.000000	0.000000	1.000	1000
100	0.01	0.15	1000	50	10	0.415313	0.584687	0.569	1000
100	0.01	0.15	1000	100	10	0.306452	0.693548	0.566	1000
100	0.01	0.15	1000	150	10	0.146572	0.853428	0.577	1000
100	0.01	0.15	1000	200	10	0.037825	0.962175	0.577	1000
100	0.01	0.15	1000	250	10	0.004415	0.995585	0.547	1000
100	0.01	0.15	1000	300	10	0.000000	1.000000	0.573	1000
100	0.01	0.15	1000	350	10	0.006579	0.993421	0.544	1000
100	0.01	0.15	1000	400	10	0.000000	1.000000	0.573	1000

Table 7: Content bias detection under time-averaged sampling ($w = 10$). Signal-of-Selection Rate (SSR), False Negative Rate (FNR), and proportion of variants excluded (%NA) from FIT across univariate parameter sweeps: innovation rate (μ), population size (N), and total timesteps (t).

Varying b									
N	μ	b	B	t	w	SSR	FNR	%NA	Runs
100	0.01	0	1000	100	10	0.019355	0.980645	0.845	1000
100	0.01	0.05	1000	100	10	0.213389	0.786611	0.761	1000
100	0.01	0.1	1000	100	10	0.385093	0.614907	0.678	1000
100	0.01	0.15	1000	100	10	0.303030	0.696970	0.571	1000
100	0.01	0.2	1000	100	10	0.126374	0.873626	0.454	1000
100	0.01	0.25	1000	100	10	0.057725	0.942275	0.411	1000
100	0.01	0.3	1000	100	10	0.008000	0.992000	0.375	1000
100	0.01	0.35	1000	100	10	0.000000	1.000000	0.298	1000
100	0.01	0.4	1000	100	10	0.000000	1.000000	0.257	1000
Varying w									
100	0.01	0.15	1000	50	5	0.710938	0.289062	0.488	1000
100	0.01	0.15	1000	100	10	0.273159	0.726841	0.579	1000
100	0.01	0.15	1000	200	20	0.009926	0.990074	0.597	1000
100	0.01	0.15	1000	400	40	0.000000	1.000000	0.583	1000
100	0.01	0.15	1000	500	50	0.000000	1.000000	0.595	1000
100	0.01	0.15	1000	700	70	0.000000	1.000000	0.586	1000
100	0.01	0.15	1000	800	80	0.000000	1.000000	0.607	1000
100	0.01	0.15	1000	1000	100	0.000000	1.000000	0.598	1000
100	0.01	0.15	1000	2000	200	0.000000	1.000000	0.588	1000

Table 8: Content bias detection under time-averaging scheme. Signal-of-Selection Rate (SSR), False Negative Rate (FNR), and proportion of variants excluded (%NA) from FIT across univariate parameter sweeps: bias strength (b) and time window width (w).

BASELINE								
N	μ	c	B	t	SSR	FNR	%NA	Runs
100	0.02	0.05	1000	10	0.210923	0.789077	0.469	100
<i>Varying c</i>								
100	0.02	0.01	1000	50	0.060606	0.939394	0.934	1000
100	0.02	0.05	1000	50	0.012658	0.987342	0.921	1000
100	0.02	0.1	1000	50	0.040000	0.960000	0.925	1000
100	0.02	0.15	1000	50	0.014085	0.985915	0.929	1000
100	0.02	0.2	1000	50	0.016949	0.983051	0.941	1000
100	0.02	0.25	1000	50	0.000000	1.000000	0.928	1000
100	0.02	0.3	1000	50	0.000000	1.000000	0.934	1000
100	0.02	0.4	1000	50	0.018182	0.981818	0.945	1000
100	0.02	0.5	1000	50	0.000000	1.000000	0.947	1000
100	0.02	0.6	1000	50	0.000000	1.000000	0.941	1000
100	0.02	0.7	1000	50	0.000000	1.000000	0.953	1000
100	0.02	0.8	1000	50	0.000000	1.000000	0.949	1000
<i>Varying $\mu \times c$</i>								
100	0.01	0.01	1000	50	0.070175	0.929825	0.943	1000
100	0.05	0.05	1000	50	0.000000	1.000000	0.215	1000
100	0.1	0.1	1000	50	0.000000	1.000000	0.001	1000
100	0.15	0.15	1000	50	0.003000	0.997000	0.000	1000
100	0.2	0.2	1000	50	0.001000	0.999000	0.000	1000
100	0.25	0.25	1000	50	0.008000	0.992000	0.000	1000
100	0.4	0.4	1000	50	0.023092	0.976908	0.004	1000
<i>Varying $N \times c$</i>								
10	0.02	0.01	1000	50	0.383333	0.616667	0.940	1000
50	0.02	0.05	1000	50	0.152174	0.847826	0.954	1000
100	0.02	0.1	1000	50	0.012658	0.987342	0.921	1000
150	0.02	0.15	1000	50	0.000000	1.000000	0.792	1000
200	0.02	0.2	1000	50	0.000000	1.000000	0.503	1000
250	0.02	0.25	1000	50	0.000000	1.000000	0.250	1000
400	0.02	0.4	1000	50	0.007092	0.992908	0.013	1000

Table 9: Conformist bias detection under snapshot sampling. Signal-of-Selection Rate (SSR), False Negative Rate (FNR), and proportion of variants excluded (%NA) from FIT across univariate and multivariate parameter sweeps: conformist bias strength (c), innovation rate (μ) and c , and population size (N) and c .

BASELINE									
N	μ	c	B	t	w	SSR	FNR	%NA	Runs
100	0.02	0.10	1000	100	10	0	1	0.05	100
<i>Varying c</i>									
100	0.02	0.01	1000	100	10	0	1	0.063	1000
100	0.02	0.025	1000	100	10	0	1	0.070	1000
100	0.02	0.05	1000	100	10	0	1	0.073	1000
100	0.02	0.075	1000	100	10	0	1	0.072	1000
100	0.02	0.10	1000	100	10	0	1	0.069	1000
100	0.02	0.125	1000	100	10	0	1	0.077	1000
100	0.02	0.15	1000	100	10	0	1	0.073	1000
100	0.02	0.175	1000	100	10	0	1	0.077	1000
100	0.02	0.20	1000	100	10	0	1	0.071	1000
100	0.02	0.225	1000	100	10	0	1	0.059	1000
100	0.02	0.25	1000	100	10	0	1	0.076	1000
100	0.02	0.275	1000	100	10	0	1	0.053	1000
<i>Varying $\mu \times c$</i>									
100	0.01	0.01	1000	400	40	0	1	0.034	1000
100	0.05	0.05	1000	400	40	0	1	0.115	1000
100	0.10	0.10	1000	400	40	0	1	0.170	1000
100	0.15	0.15	1000	400	40	0	1	0.201	1000
100	0.20	0.20	1000	400	40	0	1	0.259	1000
100	0.25	0.25	1000	400	40	0	1	0.281	1000
100	0.40	0.40	1000	400	40	0	1	0.384	1000
<i>Varying $N \times c$</i>									
10	0.02	0.01	1000	400	40	0	1	0.043	1000
50	0.02	0.05	1000	400	40	0	1	0.075	1000
100	0.02	0.10	1000	400	40	0	1	0.066	1000
150	0.02	0.15	1000	400	40	0	1	0.086	1000
200	0.02	0.20	1000	400	40	0	1	0.064	1000
250	0.02	0.25	1000	400	40	0	1	0.058	1000
400	0.02	0.40	1000	400	40	0	1	0.050	1000
<i>Varying N</i>									
10	0.02	0.10	1000	100	10	0.0217	0.978	0.08	1000
50	0.02	0.10	1000	100	10	0	1	0.05	1000
100	0.02	0.10	1000	100	10	0	1	0.05	1000
200	0.02	0.10	1000	100	10	0	1	0.10	1000
500	0.02	0.10	1000	100	10	0	1	0.03	1000
1000	0.02	0.10	1000	100	10	0	1	0.06	1000
2000	0.02	0.10	1000	100	10	0	1	0.03	1000

Table 10: Conformist bias detection under time-averaged sampling ($w = 10$ and 40). Signal-of-Selection Rate (SSR), False Negative Rate (FNR), and proportion of variants excluded (%NA) from FIT across univariate and multivariate parameter sweeps: conformist bias strength (c), innovation rate (μ) and c , and population size (N) and c .

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